

Factorization Algebras

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MA822

Abstract

MA822: Topics in Geometry and Topology was taught by [Professor Matiej Szczesny](#) in the spring 2026 semester (my junior year). The topic for this semester was an introduction to the factorization algebras of Kevin Costello and Owen Gwilliam, and we followed their book [[CG16](#), [CG20](#)], focusing on the first volume. The course was designed to be accessible to a wide range of students, and the main prerequisites were basic topology, the theory of manifolds, and Lie theory.

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Chapter 1

Introduction

Lecture 1: Syllabus day

Tue 20 Jan 2026 14:06

We will move the thursday class to 2-3:30 friday. Maybe cds424?

A factorization algebra is essentially an algebraic formulation of observables in a QFT. Examples are numerous: associative algebras, vertex algebras, $\mathbb{E}_n$ -algebras over the little n -cubes operad, etc.

FAs are of course motivated by QFT, and more generally a classical field theory. A classical field theory is a space $\mathcal{E}$ of classical fields, typically the space $\Gamma(E)$ of sections of some bundle E over a manifold M , usually a fiber bundle; and an action functional $S : \mathcal{E} \rightarrow \mathbb{R}$ or $\mathbb{C}$. Write CLFT for classical field theory. In CLFT, we look for the *critical locus*

$$\text{Crit}(S) = \left\{ \phi \in \mathcal{E} \mid \frac{\delta S}{\delta \phi} = 0, dS|_{\phi} = 0 \right\}.$$

These are solved by the Euler-Lagrange equations.

For example, classical mechanics on $\mathbb{R}^n$. Take $\mathcal{E} = \text{Hom}(I, \mathbb{R}^n)$ for $I = [0, 1]$. The category can be any function, smooth maps, piecewise linear, etc. The action functional can be defined as follows. Given $q \in \mathcal{E}$, take

$$S(q) = \frac{1}{2} \int_0^1 \langle \dot{q}(t), \dot{q}(t) \rangle dt = \frac{1}{2} \int_0^1 E(t) dt,$$

where E is the kinetic energy. Take the functional derivative of S to extremize S locally. These are of course just geodesics in $\mathbb{R}^n$, meaning they satisfy

$$\frac{d^2 q}{dt^2} = 0$$

since $\Gamma_{ij}^k = 0$ on $\mathbb{R}^n$. Of course these are functions of the form $at + b$, $a, b \in \mathbb{R}$ fixed. We also obtain another constraint from the IBP (see my ma721 project): $\langle \dot{q}, \delta q \rangle|_0^1 = 0$ for any variation δq of q . If we don't fix the endpoints, then $\delta q(0) \neq \delta q(1)$ in general. This forces $\dot{q} = 0$ at $0, 1$, and since $\dot{q} = a$ we have $a = 0$. Thus $q = b \in \mathbb{R}$ is constant. This is boring. Hence why we fix straight endpoints of q . We thus define

$$\frac{\delta S}{\delta q} = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0} S(q + \varepsilon \delta q)$$

for a variation δq of q fixing endpoints. In this case, $\text{Crit}(S) \subseteq \mathcal{E}$ is a $2n$ -dimensional space when $\mathcal{E}$ is functions on $\mathbb{R}^n$.

Definition 1.0.1. Define the *classical observables* $\mathcal{O}^{cl}$ as the algebra of functions on $\text{Crit}(S)$. This is a fairly informal definition, since we have not specified what types of functions we are considering. Sometimes we will consider all functions $\text{Set}(E, \mathbb{R})$, but since E is rarely finite-dimensional we often restrict to one of the two following important subalgebras:

- Smooth functions $C^\infty(\text{Crit}(S), \mathbb{R})$;
- Polynomials in the coefficients of $\text{Crit}(S)$ elements.

We have another example: scalar FT, which we will call scalar ϕ^m theory on $\mathbb{R}^n$. Take $\mathcal{E} = C^\infty(\mathbb{R}^n)$, and

$$S(\phi) = \int_{\mathbb{R}^n} \frac{1}{2} \phi \Delta \phi + \frac{g}{m!} \phi^m \, d\text{Vol}.$$

Here Δ is the laplacian, which exists on any Riemannian manifold. However we are integrating over something noncompact, so we will later get around this issue. Intuitively ϕ must decay sufficiently rapidly at ∞ . We get around it by taking the derivative of the action. We also only consider $\delta\phi$ that are compactly supported. Thus the functional derivative wrt $\delta\phi$ always converges. Extremitization is more complex but we obtain

$$\Delta\phi + \frac{g}{(m-1)!} \phi^{m-1} = 0.$$

If $m = 1, 2$ then this is a linear PDE, and we call it a *free field theory*. Once m is higher, the ϕ^{m-1} term is nonlinear in ϕ . If $g = 0$ then ϕ is harmonic. These are the various characterizations of solutions to special cases of this PDE, and most critical loci are not finite dimensional.

Finally, just note that even if S is not well-defined, if we only vary wrt compactly supported variations then the critical locus is well-defined.

Summary of first day: A field theory is a set $\mathcal{E}$ of fields and an action functional $S : \mathcal{E} \rightarrow \mathbb{R}$. Usually $\mathcal{E}$ is the vector space $\Gamma(E)$ of a vector bundle $\pi : E \rightarrow M$ for a manifold M , generally a fiber bundle defined below. The *critical locus* $\text{Crit}(S)$ is the set of extrema for S , also denoted $EL(S)$ for Euler-Lagrange. These are the equations of motion. Classical observables are then defined as the algebra of functions $\text{Set}(\text{Crit}(S), \mathbb{R})$, smooth functions $C^\infty(\text{Crit}(S), \mathbb{R})$, or polynomials in $\text{Crit}(S)$. The way to interpret the final one is less formal. For an example, let $\mathcal{E} = C^\infty(I, \mathbb{R}^n)$ such that $\text{Crit}(S) = \{q(t) = at + b \mid a, b \in \mathbb{R}^n\}$. Then the example is $\mathbb{R}[a^i, b^i]$ – that is, polynomials in components of the coefficients of the extrema.

Often times, $\mathcal{E}$ is not a finite-dimensional vector space, meaning to give it the structure of a smooth manifold we would need to pass to infinite-dimensional differential geometry. This is possible, but quite difficult, and we ignore this point for now, since this is not an important point.

Lecture 2: Finishing classical motivation and beginning quantum motivation

Fri 23 Jan 2026 13:59

Recall that the issue of boundary conditions vanishes by restricting to compactly supported variations.

Definition 1.0.2. Let $U \subseteq M$ open. Define $EL(U)$ as solutions to E-L eqns on U via restricting $S|_U$. Define $\mathcal{O}^{cl}(U)$ to be functions on $EL(U)$.

We are intentionally vague with what “functions on $EL(U)$ ” means. Generally the critical locus is at least topologized since we can endow $\mathcal{E}$ with the compact-open topology and consider $EL(U)$ as a subspace. The point is that we are thinking of $\mathcal{O}^{cl}$ as some sheaf-like construction. We will then axiomatize factorization algebras as such structures.

Note that $U \subseteq V$ induces $EL(V) \rightarrow EL(U)$ via restriction, which then induces $\mathcal{O}^{cl}(U) \rightarrow \mathcal{O}^{cl}(V)$ via precomposition. Thus $\mathcal{O}^{cl}$ is a *precosheaf*. We are thus interested in how $\mathcal{O}^{cl}$ interacts with open sets.

If $U_1, U_2 \subseteq M$ are open and disjoint, then specifying a solution to the Euler Lagrange eqns on $U_1 \cup U_2$ is the same as specifying a solution on U_1 then U_2 , since the solution on one region does not impact the other. Thus we have

$$EL(U_1 \amalg U_2) \cong EL(U_1) \times EL(U_2).$$

We want to see how this changes when we take $\mathcal{O}^{cl}$. To take functions on a product $EL(U_1) \times EL(U_2) \rightarrow \mathbb{R}$, we should be able to pass to the tensor product. For example, if we consider polynomials, then we can view polynomials on U_1 as polynomials in x and polynomials on U_2 as polynomials in y . We then simply note that $\mathbb{C}[x] \otimes_{\mathbb{C}} \mathbb{C}[y] \cong \mathbb{C}[x, y]$, and similarly for other fields. There is a bit of a more rigorous argument that I will not get into. The point is that for classes of functions we are considering, we should have

$$\mathcal{O}^{cl}(U_1 \amalg U_2) \cong \mathcal{O}^{cl}(U_1) \otimes \mathcal{O}^{cl}(U_2).$$

This isomorphism need not hold in maximal generality, but there are various tools we can use to make it hold more generally, such as completing a ring with respect to a topology. This is apparently discussed further in [CG16].

We should expect $\mathcal{O}$ to pass a notion of functoriality to these tensor products, meaning that given a collection $\{U_i \subseteq M\}_{i \in n}$ of pairwise disjoint open sets and an open set $V \subseteq M$, there should be a *structure map*

$$\bigotimes_{i \in n} U_i \rightarrow V.$$

It may be useful to think of this like a functor of colored operads.

In classical theory, when $U_1 \cap U_2$ is nonempty, it is straightforward to just say we consider solutions that agree on the overlap. So we can consider overlaps. However in a quantum theory, it doesn't make sense to make measurements at the same time (recall our manifold represents *spacetime*), since observables don't commute and thus measuring two things at once would not just be measuring the particle, but the other measurement would be interfering. Hence we cannot consider overlaps in quantum theory. Thus we do not have a cosheaf.

Chapter 2

From gaussian measures to factorization algebras

2.1 Gaussian integrals in finite dimensions

We now consider another motivation via Feynman's path integral and the BV formalism. Let (X, μ) a probability space (measure space with a specific measure that behaves like you would expect) and random variables $\{f_i : X \rightarrow \mathbb{C}\}_{i \in r}$. The expectation value is defined by $\langle \{f_i\} \rangle = \int_X \prod_i f_i d\mu$. For quantum mechanics, we replace classical variables with classical observables $\mathcal{E} \rightarrow \mathbb{C}$ and we replace μ with $e^{-S(\phi)} D\phi$. This yields the path integral

$$\left\langle \left\{ \mathcal{O}_i^{cl} \right\}_{i \in r} \right\rangle = \int_{\mathcal{E}} \prod_{i \in r} \mathcal{O}_i(\phi) e^{-S(\phi)} D\phi.$$

There is a problem introduced by the $D\phi$ term. This should be a measure on the space $\mathcal{E}$ of fields, but such a measure either doesn't exist, or is sufficiently poorly behaved to be wrong. For example, on topological vector spaces such a measure is not translation invariant.

Thus it's hard (for mathematicians) to figure out what this means. We can make sense of this in the perturbative case, but physicists and mathematicians agree this is not the whole picture. This is a serious problem, because physicists use the path integral to obtain correct physical answers, so there must be some rigorous mathematical definition of the path integral. This would be quite useful to find, since computing expectation values can yield nearly all the information in quantum physics.

Let's reframe integrals in homological terms. Let M a compact orientable n -manifold (compactness not required, but simpler) without boundary. Recall the de Rham complex $(\Omega^\bullet(M), d)$ is truncated at n , and yields de Rham cohomology. The top cohomology $H_{dR}^n(M) = \Omega^n(M)/d\Omega^{n-1}(M)$ is isomorphic to $\mathbb{R}$ since it is compact. Since we integrate n -forms, we can view integration as a map

$$\int : \Omega^n(M) \rightarrow \mathbb{R}.$$

By Stokes' theorem, the kernel of $\int$ is exact forms since $\partial M = \emptyset$, so it lowers by universality to

$$\int : H^n(M) \cong \mathbb{R}.$$

The issue is when passing to infinite dimensional manifolds with $M = \mathcal{E}$, where top forms don't make sense. To make sense of this, we review another characterization of the de Rham complex in the finite case.

Recall that given a k -vector field $X \in \Gamma(TM^{\otimes k})$ and an ℓ -form $\omega \in \Omega^\ell(M)$, then the interior product $X \lrcorner \omega$ is defined as the $(\ell - k)$ -form defined by

$$(X \lrcorner \omega)_p = \omega_p(X_p, -).$$

This makes sense since X_p is a k -tensor, and up to multilinearity represents k vectors. Thus $\omega(X, -)$ is a map in $\ell - k$ variables.

Recall also that a *volume form* for a Riemannian manifold is a form ω such that for every oriented orthonormal local frame $\{E_i\}_{i=1}^n$, we have $\omega(E_1, \dots, E_n) = 1$. In smooth coordinates,

$$\omega_g = \sqrt{\det g_{ij}} dx^1 \wedge \dots \wedge dx^n.$$

Take $\omega \in \Omega^n(M)$ a volume form. This is nondegenerate, so the interior product $- \lrcorner \omega$ yields isomorphisms $\Lambda^k TM \cong \Lambda^{n-k} T^*M$. Thus contracting with ω yields by functoriality of taking sections

$$\Gamma(\Lambda^{n-k} TM) \cong \Omega^k(M).$$

We call these *polyvector fields*. We can induce a differential on $\Lambda^\bullet TM$ to promote this to a map of cochain complexes:

$$\begin{array}{ccccccc} \dots & \xrightarrow{d} & \Omega^{n-1}(M) & \xrightarrow{d} & \Omega^n(M) & \xrightarrow{d} & 0 \\ & & \uparrow \cong \lrcorner \omega & & \uparrow \cong \lrcorner \omega & & \\ \dots & \xrightarrow{?} & \mathfrak{X}(M) & \xrightarrow{?} & C^\infty(M) & \xrightarrow{?} & 0 \\ & & \parallel & & \parallel & & \\ \dots & & \Gamma(\Lambda^1 TM) & & \Gamma(\Lambda^0 TM) & & \end{array}$$

We can compute the map $\mathfrak{X}(M) \rightarrow C^\infty(M)$ using the fact that contraction with ω is an isomorphism. Let $X \in \mathfrak{X}(M)$. Then the image of X in $\Omega^n(M)$ is $di_X \omega$. We then use Cartan's magic formula ([Lee00] 14.35)

$$\mathcal{L}_X \omega = i_X d\omega + di_X \omega$$

to compute

$$\mathcal{L}_X d\omega = i_X d^2 \omega + di_X \omega = di_X \omega.$$

Hence the image of X in $C^\infty(M)$ must contract with ω to yield $\mathcal{L}_X \omega$. Since contracting with a function is simply pointwise multiplication and ω is nonvanishing, it makes sense to divide by it, yielding the map

$$X \mapsto \frac{\mathcal{L}_X d\omega}{\omega} = \frac{d\mathcal{L}_X \omega}{\omega}.$$

This map turns out to be Div_ω . We call the complex the *divergence complex* and in finite dimensions it is isomorphic to the de Rham complex. The relevance is that it can pass to infinite dimensions by rephrasing integration as a map

$$\int : \Omega^n(M) \cong \Gamma(\Lambda^{n-n=0} TM) = C^\infty(M) \longrightarrow \mathbb{R}$$

and remarking that in infinite dimensions, the de Rham complex is concentrated in positive degree, whereas the divergence complex must be concentrated in negative degree. We can hope to make sense of div_ω , called the BV laplacian Δ , in ∞ dim when $M = \mathcal{E}$.

Some remarks from [CG16] that I find important. Perturbative theory corresponds to working over the ring $\mathbb{C}[[\hbar]]$ of formal power series in $\hbar$ (so like stuff is just power series). At $\hbar = 0$, we recover classical theory. In perturbative theory, the vector space $\mathcal{O}(U)$ of observables on an open set U will form a flat $\mathbb{C}[[\hbar]]$ -module.

Lecture 3: The quantum motivation

Tue 27 Jan 2026 14:01

So far we have been fairly imprecise in our motivation of (pre)factorization algebras using the classical reasoning for preFAs. Now we are exploring the *quantum* motivation for prefactorization algebras via the path integral. We will for now think of observables $\mathcal{O}^{cl}$ as functions on *all fields*, not just the critical locus. Recall that we want something like

$$\langle \mathcal{O}_1^{cl}, \dots, \mathcal{O}_r^{cl} \rangle = \int_{\phi \in \mathcal{E}} \mathcal{O}_1^{cl}(\phi) \cdots \mathcal{O}_r^{cl}(\phi) e^{-\frac{1}{\hbar} S(\phi)} \mathcal{D}\phi.$$

Of course this should be some kind of probability measure, but it doesn't really make mathematical sense. Recall that we invoked the isomorphism of forms and polyvector fields to construct the *divergence complex*, provided we have a volume form. The maps $\Gamma(\Lambda^{n-k} TM) \rightarrow \Gamma(\Lambda^{n-k-1} TM)$ are the divergence maps. In de Rham, integration lowers to

$$\int : H^n(M) \cong \mathbb{R}.$$

Thus we can identify along the chain isomorphism to think of the projection $C^\infty(M) \rightarrow C^\infty(M)/\text{Im div}_\omega$ as $f \mapsto \int_M f \omega = \langle f \rangle$.

Let's derive prefactorization algebras from this. Consider the free scalar field theory $\mathcal{E} = C^\infty(\mathbb{R}^n)$,

$$S(\phi) = \int_{\mathbb{R}^n} \phi(\Delta + m^2) \phi \, d^n x.$$

Although this need not make sense, recall that provided our variations are compactly supported, extremizing S does indeed make sense. Fix A a positive definite $n \times n$ symmetric matrix. Write $\mathbb{P}(\mathbb{R}^n) = \mathbb{R}[x_1, \dots, x_n]$. Then recall a Gaussian integral is of the form $\langle - \rangle : \mathbb{P}(\mathbb{R}^n) \rightarrow \mathbb{R}$ by

$$f \mapsto \int_{\mathbb{R}^n} f(x) \exp\left(-\frac{1}{2} x^t A x\right).$$

This makes sense since A is positive definite. We claim that path integrals

$$\int_{\mathcal{E}} \mathcal{O}_1^{cl}(\phi) \cdots \mathcal{O}_r^{cl} \exp\left(\int -\frac{1}{2} \phi(\Delta + m^2) \phi \, d^n x\right)$$

are like an infinite dimensional version of the first problem. This follows since

$$\phi \otimes \psi \mapsto \int \frac{1}{2} \phi(\Delta + m^2) \psi \, d^n x$$

is positive definite and symmetric. Oh btw m is mass.

Now let's try to generalize. Consider $\mathbb{R}^n$ which is noncompact. We have $\text{div}_\omega : \mathbb{P}\mathfrak{X}(\mathbb{R}^n) \rightarrow \mathbb{P}(\mathbb{R}^n)$ and then integration maps to $\mathbb{R}$. We cannot consider all of C^∞ since $\mathbb{R}^n$ is contractible so its top de Rham cohomology is 0. bla bla see picture i aint typin allat rn

Now we want to integrate over $C^\infty(\mathbb{R}^n)$ rather than just $\mathbb{R}^n$. We should start by defining polynomials on $V = C^\infty(\mathbb{R}^n)$. Given a finite vector space W , a polynomial on W is a symmetric algebra on W^* . This is since stuff like x_i are projections (lin functionals) so $x_1^2 x_2 \cdots$ something like that is a symmetric product of those functionals. Hence motivating a polynomial.

We don't want the entire algebraic dual of V , since it would be massive. It would make more sense to at least restrict to *continuous* functionals with respect to some natural topology on $C^\infty(\mathbb{R}^n)$,

like the compact-open topology. Certainly we want the continuous functions ev_x evaluating at x . We can rewrite them as distributions using the dirac delta (a distribution)

$$\text{ev}_x f = \int_{\mathbb{R}^n} f(z) \delta(z - x) \, d^n z.$$

Thus the correct analogue of polynomials here is something like the symmetric algebra on the space of distributions on $\mathbb{R}^n$. Integrating against compactly supported functions yields an inclusion of compactly supported C^∞ functions into distributions. We will restrict to those for our discussion. Delta functions and evaluation functionals are not in the image of this inclusion. We will write this as $C_c^\infty(\mathbb{R}^n)$, and note that this is dense in the space of distributions. For our purposes, distributions are defined as dual to a space of functions.

Apparently distributions generalize functions and it makes sense to consider diffeqs of distributions, so if you can't find a function that satisfies your PDE then you can look for a distribution.

We define $\mathbb{P}(C^\infty(\mathbb{R}^n)) := \text{Sym } C_c^\infty(\mathbb{R}^n)$.

Lecture 4: Formalizing Gaussian integrals over infinite dimensional manifolds

Fri 30 Jan 2026 14:03

For now we will continue to meet in psy b33 pending further room changes.

First recall some stuff from the previous lecture. A gaussian integral is of the form

$$\int_{x \in \mathbb{R}^n} \exp \left(-\frac{1}{2} \sum_{i,j} x_i A_{ij} x_j \right) \, d^n x$$

for A positive-definite and symmetric. Here positive definite means $x^t A x > 0$ for all nonzero $x \in \mathbb{R}^n$. These integrals are analogous to our definition of the path integral in the following way. By recognizing the exponential as $-1/2 x^t A x$, by changing $x \rightarrow \phi$ and $A \rightarrow (\Delta + m^2)$, we obtain something like the free ϕ -field theory path integral. In quantum mechanics, it is standard to consider a combinatorial approach to this integral, and to generalize it to infinite dimensions. We instead focus on the divergence operator we previously defined.

Let $\omega = \exp \left(-\frac{1}{2} \sum_{i,j} x_i A_{ij} x_j \right) \, d^n x$ be a volume form (aka a measure) on $M = \mathbb{R}^n$. Recall that we are defining the divergence operator abstractly via

$$\text{Div}_\omega X \omega = \mathcal{L}_X \omega$$

for any $X \in \mathfrak{X}(M)$. Then for $X = \sum_i f_i \partial_i$, we can compute using Cartan's magic formula that

$$\text{Div}_\omega X = \text{Div}_\omega \left(\sum_i f_i \partial_i \right) = - \sum_{i,j} f_i A_{ij} x_j + \sum_i \partial_i f_i.$$

It's convenient to restrict only to polynomial $P(\mathbb{R}^n)$ on $\mathbb{R}^n$. By diagonalizing A , it follows that the kernel of Div_ω has codimension 1 in $P(\mathbb{R}^n)$, and thus it lowers to an isomorphism with $\mathbb{R}$, as desired. This justifies working in polynomials, since we may once again identify integration with computing the divergence operator in homology.

Now we pass to infinite dimensional manifolds. Such a measure ω no longer exists, so our task is to generalize our definition of divergence in such a way that makes sense of a "measure" of the form

$$\tilde{\omega} = \exp \left(\int_{\mathbb{R}^n} -\frac{1}{2} \phi (\Delta + m^2) \phi \, d^n x \right) D\phi.$$

Taking inspiration from $\mathbb{R}^n$, we try to work in polynomials once again. It is useful also to consider the set $\text{Vect}(\mathbb{R}^n)$ of *polynomial vector fields* on $\mathbb{R}^n$, meaning for $X \in \text{Vect}(\mathbb{R}^n)$, there is a coordinate representation of X as

$$X = \sum_i P_i(x) \frac{\partial}{\partial x_i},$$

for $P_i \in \mathbb{R}[x_1, \dots, x_n]$ a polynomial on $\mathbb{R}^n$.

In order to begin replicating this process in the infinite dimensional case of $C^\infty(U)$, we need to define polynomials in $C^\infty(U)$. We can do this slightly more generally. Let V a vector space, and note that $C^\infty(U)$ is a vector space, hence this generalizes our situation. We cannot directly define polynomials on V , but we *can* pass to the symmetric algebra $\text{Sym}(V)$. However, this is a somewhat naïve approach. A *monomial* on $\mathbb{R}^n$ is a linear functional on $\mathbb{R}^n$; it is precisely an element of $\text{Vect}_{\mathbb{R}}(\mathbb{R}^n, \mathbb{R})$. Hence we would expect that polynomials would instead be defined as something like $\text{Sym}(V^*)$.

This is overkill for our approach. If we only want to look at $V = C^\infty(U)$, then the dual space of this can be identified with the set of *compactly supported distributions* $\mathcal{D}_c(U)$ in U . This is a bit more than we need, so we note that we can identify the set $C_c^\infty(U)$ of compactly supported smooth functions on U with a subspace of $\mathcal{D}_c(U)$ via, for all $g \in C_c^\infty(U)$, considering $g \in \mathcal{D}_c(U)$ as the linear functional

$$g(\phi) = \int_U g \phi \, d^n x.$$

It can be shown $C_c^\infty(U)$ is dense in the set of distributions, so functions such as the Dirac delta admit approximations in $C_c^\infty(U)$. Hence we work with $C_c^\infty(U)$, since it is convenient for carrying out the calculus of variations.

With this understanding, a priori we might take polynomials to be $\tilde{P}(C^\infty(U)) = \text{Sym}(C_c^\infty(U))$ the symmetric algebra. This makes intuitive sense, since a monomial in n variables would be some coefficient times an n -fold product in linear functionals, hence an element of the n th symmetric power. A full polynomial would then be a finite linear combination of elements of the symmetric algebra. We will see issues with this definition, hence the tilde. However it is good motivation.

From this, we note that given V a vector space, there is an identification $T_p V \cong V$. Since a polynomial vector field is a linear combination in tangent vectors with polynomial coefficients, we might a priori take

$$\widetilde{\text{Vect}}(C^\infty(U)) := \text{Sym}(C_c^\infty(U)) \otimes C^\infty(U) \cong \text{Sym}(C_c^\infty(U)) \otimes T_p C^\infty(U).$$

Here we recall that $\tilde{P}(C^\infty(U)) := \text{Sym}(C_c^\infty(U))$.

Let $\{f_i\}_{1 \leq i \leq r} \subseteq C_c^\infty(U)$ such that their product $\prod_i f_i$ is a monomial in $P(C^\infty(U))$. Then such a monomial acts on $C^\infty(U)$ via

$$\begin{aligned} \left(\prod_i f_i \right) (\phi) &= \int_{x \in U^r} \prod_i f_i(x_i) \phi(x_i) \, d\text{vol}(x_1) \wedge \dots \wedge d\text{vol}(x_r) \\ &= \prod_i \int_U f_i \phi \, d\text{vol}. \end{aligned}$$

Note that this is the same as the usual action, simply generalized in the natural way for symmetric products. Imposing linearity allows arbitrary polynomials to act on $C^\infty(U)$.

An element of $\widetilde{\text{Vect}}(C^\infty(U))$ is a finite linear combination of the form $X = f_1 \dots f_r \partial_\phi$ with $f_i, \phi \in C_c^\infty(U)$. [What is ∂_ϕ ?] Now let $g_1 \dots g_m$ be a monomial in $P(C^\infty(U))$. Then X can act on

$g_1 \cdots g_s$ by imposing linearity, the product rule, and applying the action of polynomials on $C^\infty(U)$:

$$f_1 \cdots f_r \frac{\partial}{\partial \phi}(g_1 \cdots g_s) = f_1 \cdots f_r \sum_{i=1}^s g_1 \cdots \widehat{g_i} \cdots g_s \int_U g_i \phi \, d\text{vol}.$$

Definition 2.1.1. The divergence operator associated to the action functional

$$S(\phi) = \int_U \phi(\Delta + m^2) \phi \, d\text{vol}$$

is the linear map

$$\widetilde{\text{Div}}_\omega : \widetilde{\text{Vect}}(C^\infty(U)) \rightarrow \widetilde{P}(C^\infty(U))$$

defined by, for $f_i, \phi \in C_c^\infty(U)$,

$$\widetilde{\text{Div}}_\omega \left(f_1 \cdots f_r \frac{\partial}{\partial \phi} \right) = -f_1 \cdots f_r (\Delta + m^2) \phi + \sum_{i=1}^r f_1 \cdots \widehat{f_i} \cdots f_r \int_U \phi f_i \, d\text{vol}.$$

Now this is generally not sufficient. Later in the text, we will see an issue with making homological algebra work with topological vector spaces. It is generally difficult to do homological algebra without breaking the topological properties of the vector spaces. The solution we will employ is working with a certain category that we will not discuss in class, but is in the book. It sounds like some kind of Grothendieck construction. To work in this category, we need to consider a *completed* tensor product.

Note that there is a canonical map $C^\infty(M) \otimes C^\infty(M) \rightarrow C^\infty(M \times M)$ by $f \otimes g \mapsto f(x)g(y)$; x coordinates on the first M and y on the second. This is injective but not surjective, but the image is dense. We define $C^\infty(M \times M)$ to then be the *completed* tensor product of $C^\infty(M)$ with itself. Moreover note that C^∞ is lax monoidal.

In a similar manner, we define the *completed* symmetric power $\text{Sym}^n(C^\infty(M))$ to be $C^\infty(M^n)^{S_n}$, which comes with an action of S_n , which it also injects into. This means we need to modify our previous definitions to introduce the completed tensor and symmetric power. Then

$$\mathbb{P}(C^\infty(V)) = \bigoplus_{n \geq 0} C_c^\infty(V^n)^{S_n}$$

$$\text{Vect}(C^\infty(V)) = \bigoplus_{n \geq 0} C_c^\infty(V^{n+1})^{S_n}.$$

Here the $n+1$ is because we want an extra component for the vector part. So now $\widetilde{\text{Div}}$ extends to the completion

$$\text{Vect}_C(C^\infty(V)) \mapsto F(\Delta + m^2)_{x_{n+1}} F(x_1, \dots, x_{n+1}) - \sum_{i=1}^n \int_{x_i \in U} F(x_1, \dots, x_i, \dots, x_n, x_i) \, d\text{vol}.$$

Definition 2.1.2. Quantum observables $\text{Obs}^q(U)$ on $U \subseteq M$ are defined as the cokernel of divergence:

$$\text{Obs}^q(U) := \frac{\mathbb{P}(C^\infty(U))}{\text{Im Div}} \cong \text{Coker Div}.$$

We think of quantum observables as path integrals of fields supported in U . This makes sense due to our recharacterization of integration via de Rham cohomology and the divergence complex. We now want to know what the structure of this is for different open sets $U, V \subseteq M$?

Polynomials, even in infinite dimensions, form a commutative ring. Thus $\mathbb{P}(C^\infty(U))$ has a commutative ring structure. Write $\mathbb{P}^n(C^\infty(U)) := C_c^\infty(U^n)^{S_n}$ such that $\mathbb{P}(C^\infty(U)) = \bigoplus \mathbb{P}^n(C^\infty(U))$. Then we can define

$$\mathbb{P}^n(C^\infty(U)) \otimes \mathbb{P}^m(C^\infty(U)) \rightarrow \mathbb{P}^{n+m}(C^\infty(U)).$$

This follows by symmetrizing the product of two polynomials

$$F(x_1, \dots, x_n) \otimes G(x_{n+1}, \dots, x_{n+m}) \mapsto \frac{1}{|S_{m+n}| = (m+n)!} \sum_{\sigma \in S_{n+m}} F(x_{\sigma(1)}, \dots, x_{\sigma(n)}) G(x_{\sigma(n+1)}, \dots, x_{\sigma(n+m)}).$$

Note that under inclusions $U \subseteq V$ of open sets, a compactly supported function on U is compactly supported on V , so C_c^∞ is a cosheaf. So given $F \in P(C^\infty(U))$, we can extend it to a polynomial on $P(C^\infty(V))$.

Lemma 2.1.3 ([CG16] 2.3.1). *The product map*

$$\mathbb{P}(C^\infty(U)) \otimes \mathbb{P}(C^\infty(U)) \rightarrow \mathbb{P}(C^\infty(U))$$

does not descend to the quotient, meaning it does not induce $\text{Obs}^q(U) \otimes \text{Obs}^q(U) \rightarrow \text{Obs}^q(U)$. However, if U_1, U_2 are disjoint open subsets of an open set V , then the product map

$$\mathbb{P}(C^\infty(U_1)) \otimes \mathbb{P}(C^\infty(U_2)) \rightarrow \mathbb{P}(C^\infty(V))$$

defined by composing the inclusions into $\mathbb{P}(C^\infty(V))$ with the product map does descent to such a map on quantum observables.

Recall $\text{Obs}^q(U) = \mathbb{P}(C^\infty(U)) / \text{Im Div}$. The non-existence follows since Div need not be a ring homomorphism, so the image need not be an ideal. We complete the proof next time, showing we obtain a prefactorization structure.

Lecture 5: asdbkfdjs

Tue 03 Feb 2026 13:58

Proof. Suppose we have vector spaces M, N , and subspaces $K \subseteq M$ and $L \subseteq N$. Recall that $F : M \rightarrow N$ descends to $M/K \rightarrow N/L$ whenever $F(K) \subseteq L$. We apply this to the case $P(C^\infty(U_1)) \otimes P(C^\infty(U_2)) \rightarrow P(C^\infty(V))$. The tensor product admits a projection q_{12} to $\text{Obs}^q(U_1) \otimes \text{Obs}^q(U_2)$. We can then write

$$\text{Ker } q_{12} = (\text{Im}(\text{Div}_{U_1}) \otimes P(C^\infty(U_2)) + (P(C^\infty(U_1)) \otimes (\text{Im } \text{Div}_{U_2})).$$

So we want to induce a map $\tilde{\pi}$ in the diagram

$$\begin{array}{ccccc} \text{Ker } q_{12} & \hookrightarrow & P(C^\infty(U_1)) \otimes P(C^\infty(U_2)) & \xrightarrow{q_{12}} & \text{Obs}^q(U_1) \otimes \text{Obs}^q(U_2) \\ & & \downarrow \pi & & \downarrow \pi \\ \text{Im Div}_V & \hookrightarrow & P(C^\infty(V)) & \xrightarrow{q} & \text{Obs}^q(V) \end{array}$$

It's enough to check that π identifies the kernel into the image of Im Div_V ; we may freely split the kernel at the $+$.

There is a key identity. Let X a vector field on N , f a function on N , and ω a measure. Then $\text{Div}_\omega(fX) - f \text{Div}_\omega(X) = Xf$. To prove this (in finite dimensions), recall that $\text{Div}_\omega X = \mathcal{L}_X \omega / \omega$. Then Cartan's magic formula and the product rule yield (since ω is a top form and $d\omega = 0$)

$$\begin{aligned} \text{Div}_\omega X &= \frac{d\iota_X \omega}{\omega} - f \frac{d\iota_X \omega}{\omega} = \frac{d(f\iota_X \omega)}{\omega} - f \frac{d\iota_X \omega}{\omega} \\ &= \frac{df \cdot \iota_X \omega}{\omega} + \frac{f \cdot d\iota_X \omega}{\omega} - f \frac{d\iota_X \omega}{\omega} = \frac{df \cdot \iota_X \omega}{\omega}. \end{aligned}$$

Passing to coordinates, let $\omega = g(x_i)dx^i$ and $X = h_j \partial_j$. Then

$$\iota_X \omega = \sum_{j=1}^n (-1)^j h_j g dx^1 \wedge \cdots \wedge \widehat{dx^j} \wedge \cdots \wedge dx^n.$$

Then

$$df \wedge \iota_X \omega = \sum h_j g \frac{\partial f}{\partial x_j} dx^1 \wedge \cdots \wedge dx^n.$$

Then $df \cdot \iota_X \omega / \omega$ is simply $\sum_j h_j \partial_j f = X(f)$. This proof used finite dimensions, but the statement still follows for infinite dimensional manifolds. Hence if $X \in \text{Vect}(C^\infty(U))$ and $F \in P(C^\infty(U))$, then $\text{Div}(FX) - F \text{Div}(X) = X(F)$.

We will prove the statement in the dense subspaces $\tilde{P}$ and $\widehat{\text{Vect}}$ since it is moer clear why this follows. Take $F = f_1 \cdots f_r$ and $X = g_1 \cdots g_s \frac{\partial}{\partial \phi}$, with $f_i \in C_c^\infty(U_1)$ and $g_i, \phi \in C_c^\infty(U_2)$. Then

$$X(F) = \sum_{j=1}^r g_1 \cdots g_s f_1 \cdots \widehat{f_j} \cdots f_r \int_V f_j \phi d\text{vol}.$$

But since f_j, ϕ have disjoint supports, the integral vanishes. Thus $X(F) = 0$, showing that $\text{Div}(FX) = F \text{Div}(X)$. This implies that if something is a divergence then it all is???

If divergence were an ideal in general, then Xf would always be a divergence, which need not hold. ■

This gives Obs^q the structure of a precosheaf, together with maps

$$\{U_i \subseteq V\}_{1 \leq i \leq r} \text{ disjoint} : \quad \bigotimes_{i=1}^r \text{Obs}^q(U_i) \rightarrow V.$$

Hence Obs^q is a prefactorization algebra.

In physics, there is a parameter $\hbar$ which determines the strength of quantum effects. When $\hbar = 0$, the system is classical, but for nonzero $\hbar$ (as is reality) we see quantum effects. In finite dimensions (purely for illustration), we can take a Gaussian measure

$$e^{-S/\hbar} d^n x.$$

Then

$$\text{Div}_\omega \left(\sum_i f_i \partial_i \right) = -\frac{1}{\hbar} \sum_{i=1}^n f_i \frac{\partial S}{\partial x_i} + \sum_{i=1}^n \frac{\partial f_i}{\partial x_i}.$$

Hence polynomials will simply be polynomials mod the image of this divergence, and when $\hbar \neq 0$, this is the same as the image of $\hbar \text{Div}$. This allows us to clear the denominator and take $\hbar \rightarrow 0$:

$$\text{Div}_\omega \left(\sum_i f_i \partial_i \right) = - \sum_{i=1}^n f_i \frac{\partial S}{\partial x_i} + \hbar \sum_{i=1}^n \frac{\partial f_i}{\partial x_i} \xrightarrow{\hbar \rightarrow 0} - \sum_{i=1}^n f_i \frac{\partial S}{\partial x_i}.$$

The set of all $-\sum_i f_i \partial_i S$ is the ideal of $P(\mathbb{R}^n)$ generated by partials of S . Hence in classical mechanics in finite dimensions, quantum observables go to

$$P(\mathbb{R}^n) / \langle \partial_i S \rangle_{1 \leq i \leq n}.$$

This is called the Jacobian ideal. Recall that although quantum mechanics is about path integrals (polynomials modulo divergences), classical mechanics was about critical loci. Note that by setting $\partial_i S = 0$, we obtain only elements of the critical locus of S . Hence $P(\mathbb{R}^n) / \langle \partial_i S \rangle$ is polynomial functions on the critical locus of S , meaning we recover $\mathcal{O}^{cl}(\text{Crit}(S))$. The same follows for infinite dimensional free field theories. I will not explain it because I'm tired of typing, but the various analogies of the infinite-dimensional setting to Gaussian measures should make the analogue clear.

This is a very simple example, and oftentimes what we will get is not very nice. Later on we will improve our construction by taking a differential graded object as a resolution of observables, usually a chain complex, whose zeroth cohomology is observables. In the next chapter, we will properly define prefactorization algebras, and show that there is a family $\mathcal{O}_\hbar$ of flat prefactorization algebras over $\mathbb{C}[\hbar]$, such that when $\hbar = 1$ we get quantum observables, and in $\hbar = 0$ we get classical observables.

Our permanent friday room (except one weird day in March) is CDS 548. Hopefully we have

Chapter 3

Prefactorization algebras and examples

3.1 Prefactorization algebras

Lecture 6: Prefactorization algebras

Fri 06 Feb 2026 14:05

Definition 3.1.1. A prefactorization algebra on a topological space X (usually a manifold) valued in vector spaces is a functor $\mathcal{F} : \text{Open}(X) \rightarrow \text{Vect}_k$ such that for any collection $\{U_i \subseteq V\}_{1 \leq i \leq n} \subseteq \text{Open}(X)$ of pairwise disjoint open sets, where V is open, there is a map

$$\bigotimes_{i=1}^n \mathcal{F}(U_i) \rightarrow \mathcal{F}(V)$$

which are compatible in the obvious way: let $\left\{U_i^j \subseteq V^j \subseteq W\right\}_{\substack{1 \leq j \leq m \\ 1 \leq i \leq n}}$ be collections of opens. Then the diagram

$$\begin{array}{ccc} \bigotimes_{i=1}^n \bigotimes_{j=1}^m \mathcal{F}(U_i^j) & \xrightarrow{\quad} & \bigotimes_{j=1}^m \mathcal{F}(V^j) \\ & \searrow \quad \swarrow & \\ & W & \end{array}$$

commutes.

Example 3.1.2. Any (unital) associative algebra A yields a prefactorization algebra on $\mathbb{R}$ denoted A^{fact} . Note that any open subset of the reals is at most a *countable* union of open intervals (use countability and density of $\mathbb{Q}$ to prove). To each open interval (a, b) , define $A^{fact}(a, b) = A$ with compatibility on intervals given by the identity. To any open set $U = \coprod_j I_j$, we want to set $A^{fact}(U) = \bigotimes_j A_j$, where $A_j = A$. The structure maps are simply multiplication. The issue is when the tensor is countable, this need not be defined. To do this, we require that in a countable tensor, all but finitely many elements of each tensor are the unit 1 of the algebra. Thus multiplication only needs to be performed finitely many times. The compatibility condition comes from the strict associativity of the multiplication maps. Since A need not be commutative, the ordering of the intervals in $\mathbb{R}$ yields the ordering of our elements. Raising the dimension of $\mathbb{R}^n$ can yield the notion of an $\mathbb{E}_n$ algebra here.

We can also consider the other direction, but we need to require our factorization algebra be locally constant. Inclusions of smaller intervals into bigger ones yield the identity map, but for example taking the symmetric algebra of a precosheaf does not yield this (note $\text{Sym}(\mathcal{F}(U_1) \oplus \mathcal{F}(U_2)) \cong \text{Sym}(\mathcal{F}(U_1)) \otimes \text{Sym}(\mathcal{F}(U_2))$).

Definition 3.1.3. Let $\mathcal{F}$ be a prefactorization algebra in $\mathbb{R}$ valued in vector spaces. We say $\mathcal{F}$ is locally constant if $\mathcal{F}(U) \rightarrow \mathcal{F}(V)$ is an isomorphism when $U \subseteq V$ is an inclusion of connected open intervals (more generally, a homotopy equivalence).

Lemma 3.1.4. Let $\mathcal{F}$ be a locally constant prefactorization algebra on $\mathbb{R}$ valued in Vect_k . Set $A := \mathcal{F}(\mathbb{R})$. Then $\mathcal{F} \cong A^{\text{fact}}$.

Proof. Let $I = (a, b) \subseteq \mathbb{R}$ an interval. Then $\mathcal{F}(I) \rightarrow \mathcal{F}(\mathbb{R})$ is an isomorphism. Hence all open intervals map under $\mathcal{F}$ to be isomorphic to A . We define a multiplication map for A by taking intervals $(a, b), (c, d) \subseteq I$ and considering the structure map $m : \mathcal{F}(a, b) \otimes \mathcal{F}(c, d) \rightarrow \mathcal{F}(I) \cong A$. To show A is associative, we simply apply the compatibility condition for 3 intervals. ■

We can see how (pre)factorization algebras may be used to give rise to algebras with much less trivial properties, such as vertex algebras.

Let A, B be R -algebras. We can define an (A, B) -bimodule as an R -module with compatible left- and right-actions by A, B .

Lecture 7: Further examples and a crash course in Lie theory

Tue 10 Feb 2026 14:00

MA822 will be on thursday this week from 2-3:15, and the room is still to be found; probably CDS 424.

Remark 3.1.5. If we instead take preFAs valued in chain complexes and define a locally constant preFA to preserve weak equivalences, then we get A_∞ -algebras.

Recall M from the last lecture. Define a pFA as follows. Fix $p \in \mathbb{R}$, and let U_1 denote any interval contained fully within $\mathbb{R}_{<p}$; U_2 denote any interval contained within $\mathbb{R}_{>p}$; and U_3 any interval containing p . Define $\mathcal{F}(U_1) = A$, $\mathcal{F}(U_2) = B$, and $\mathcal{F}(U_3) = M$. Let V be any interval containing the relevant subsets. The map $\mathcal{F}(U_1) \otimes \mathcal{F}(U_2) \rightarrow \mathcal{F}(V)$ is then $A \otimes B \rightarrow M$, which would be the left action of A on M . Similarly, $\mathcal{F}(U_2) \otimes \mathcal{F}(U_3) \rightarrow \mathcal{F}(V)$ is the right action $M \otimes B \rightarrow M$. The map $U_1 \subseteq V \ni p$ must be $A \rightarrow M$, and thus will be determined by $1 \mapsto m_0 \in M$. This means M should be a pointed bimodule, which is recieved from the empty set. All other maps are the identity. Coherence arises via compatibility and associativity properties. This is a special case of a *stratified* factorization algebra. In physics, this is a simple example of a *defect*: some notion of two different theories (like different potentials) that are somehow coupled at some point p .

Remark 3.1.6. The prefactorization algebra axioms imply that $\mathcal{F}(\emptyset)$ is a unital commutative algebra.

Final example. Let A, B, C unital R -algebras, M an (A, B) -bimodule, and N a (B, C) -bimodule. We can do the same thing with $M \sim p$ and $N \sim q$, but we need interaction between M and N . If $p, q \in V$, then we take $\mathcal{F}(V) = M \otimes_B N$. Here recall that algebras are rings together with an action of another ring, so this makes sense. There exist maps $M \hookrightarrow M \otimes_B N$ and $N \hookrightarrow M \otimes_B N$ by making use of the pointed-ness of M, N and mapping $m \mapsto m \otimes n_0$ and $n \mapsto m_0 \otimes n$. Most generally, $(a, m, b, n, c) \mapsto amb \otimes nc$; the b can travel between the two tensors since we tensor over $B \ni b$.

Crash course in Lie theory:

Definition 3.1.7. Let $\mathfrak{g}$ a Lie algebra. It's universal enveloping algebra $\mathcal{U}(\mathfrak{g})$ is an associative algebra which is universal in the following sense: let A a unital associative algebra and $\mathfrak{g} \rightarrow A$ a Lie algebra hom (associative algebras also have a commutator). Then there is a unique unital algebra homomorphism $\mathcal{U}(\mathfrak{g}) \rightarrow A$ making the relevant diagram commute.

Let $\text{Lie} : \text{Ass} \rightarrow \text{Lie}$ the inclusion functor and $\mathcal{U}$ the enveloping algebra functor. Then $\mathcal{U} \dashv \text{Lie}$. We construct $\mathcal{U}$ explicitly using the tensor algebra $T(\mathfrak{g})$:

$$\mathcal{U}(\mathfrak{g}) = \frac{T(\mathfrak{g})}{\langle x \otimes y - y \otimes x - [x, y] \rangle}.$$

Recall a $\mathfrak{g}$ -module is a map $\mathfrak{g} \otimes M \rightarrow M$ which is compatible with the Lie bracket: $x(ym) - y(xm) = [x, y]m$. This is precisely the same as a $\mathcal{U}(\mathfrak{g})$ -module.

Note that $\text{Mod}_{\mathfrak{g}}$ is an abelian category, so we can do (co)homology. Consider a functor $\text{inv} : \text{Mod}_{\mathfrak{g}} \rightarrow \text{Vect}_{\mathbb{C}}$ which maps M to its subspace of *invariants*: $M^{\mathfrak{g}} = \{m \in M \mid xm = 0 \forall x \in \mathfrak{g}\}$. The functor of *coinvariants* $\text{coinv} : \text{Mod}_{\mathfrak{g}} \rightarrow \text{Vect}_{\mathbb{C}}$ is the map $M \mapsto M_{\mathfrak{g}} := M/\mathfrak{g}M$.

We will reframe these in terms of $\mathcal{U}(\mathfrak{g})$ to show they are not exact. We could write $U^{\mathfrak{g}} = \text{Hom}_{\mathcal{U}(\mathfrak{g})}(\mathbb{C}, M)$. Hence $\text{inv} \cong \text{Hom}(\mathbb{C}, -)$, and is thus left-exact. We can then write $M_{\mathfrak{g}} = \mathbb{C} \otimes_{\mathcal{U}(\mathfrak{g})} M$. This follows since $a \otimes xm = ax \otimes m = 0$ for any a in $M^{\mathfrak{g}}$. Idk this one is weird. The derived functors of Hom are Ext , and the derived functors of $\otimes$ are Tor .

Definition 3.1.8. We define $H^n(\mathfrak{g}, M) = R^n \text{Mod}_{\mathcal{U}(\mathfrak{g})}(\mathbb{C}, M) = \text{Ext}_{\mathcal{U}(\mathfrak{g})}^n(\mathbb{C}, M)$. We similarly define $H_n(\mathfrak{g}, M) = L^n(\mathbb{C} \otimes_{\mathcal{U}(\mathfrak{g})} M) = \text{Tor}_{\mathcal{U}(\mathfrak{g})}^n(\mathbb{C}, M)$.

Lecture 8: Lol brian taught class tday

Thu 12 Feb 2026 14:01

Let $\mathfrak{g}$ a Lie algebra and M a $\mathfrak{g}$ -module; equivalently a $\mathcal{U}(\mathfrak{g})$ -module as a module over an associative algebra. In particular $\mathcal{U}$ induces an equivalence $\text{Mod}_{\mathfrak{g}} \cong \text{Mod}_{\mathcal{U}(\mathfrak{g})}$.

To compute the cohomology $H^i(\mathfrak{g}; M)$, we use the Chevalley-Eilenberg resolution to find a cochain complex in vector spaces whose cohomology is $H^i(\mathfrak{g}; M)$. We start by finding a free resolution of $\mathbb{C}$ as a $\mathcal{U}(\mathfrak{g})$ -module. We do this by working in dg Lie algebras.

Definition 3.1.9 (dg Lie algebra). A dg Lie algebra is a graded Lie algebra $\mathfrak{g} = \bigoplus_i \mathfrak{g}_i$ (so it has a bracket) together with a differential $d : \mathfrak{g}_i \rightarrow \mathfrak{g}_{i-1}$, where

$$[-, -] : \mathfrak{g}^i \times \mathfrak{g}^j \rightarrow \mathfrak{g}^{i+j}$$

and the bracket satisfies

$$[x, y] = -(-1)^{|x||y|}[y, x].$$

Abs value is the degree here. Also d is a derivation for the bracket:

$$d[x, y] = [dx, y] + (-1)^{|x|}[x, dy].$$

For an example, let $\mathfrak{g}$ be an ordinary Lie algebra. If we take a dg Lie algebra to be concentrated in degree 0, then $\mathfrak{g}$ is a dg Lie algebra. Define $\mathfrak{g}_{\text{dR}}$ to be the following Lie algebra (the de Rham dg Lie algebra) as the cochain complex with $\mathfrak{g}$ in degrees -1 and 0 , and vanishing everywhere else. The map is the identity. The bracket is defined as follows. Take $x, y \in \mathfrak{g}_{\text{dR}}^0$ and $x', y' \in \mathfrak{g}_{\text{dR}}^{-1}$. A general element is of the form $x + x', y + y'$. The bracket is

$$[x + x', y + y'] = [x, y] + [x, y'] + [x', y].$$

A quasi-isomorphism of dg Lie algebras is a map of dg Lie algebras which is a quasi-isomorphism when viewed as a map of cochains.

Remark 3.1.10. There is a quasi-isomorphism $\mathfrak{g}_{\text{dR}} \simeq 0$.

Lemma 3.1.11. *The universal enveloping algebra preserves cohomology.*

Theorem 3.1.12 (PBW). *As vector spaces, $\mathcal{U}(\mathfrak{g}) \cong \text{Sym}(\mathfrak{g})$. If passing to dg Lie algebras, the result holds for Sym the graded symmetric algebra on a graded vector space, which is worth googling.*

An example of the graded symmetric algebra on W a $\mathbb{C}$ -vector space is $\mathbb{C}$ in degree 0, and $\Lambda^n W$ in degree $-n$.

Lemma 3.1.13. $\mathcal{U}(\mathfrak{g}_{\text{dR}})$ is a free $\mathcal{U}(\mathfrak{g})$ -module.

Proof. The universal enveloping algebra is monoidal $(\text{Mod}_{\mathfrak{g}}, \oplus) \rightarrow (\text{Mod}_{\mathcal{U}(\mathfrak{g})}, \otimes_{\mathbb{C}})$. Additionally, $\mathcal{U}(\mathfrak{g} \oplus \mathfrak{h}) \cong \mathcal{U}(\mathfrak{g}) \otimes_{\mathbb{C}} \mathcal{U}(\mathfrak{h})$ is free (take a basis for $\mathfrak{h}$). Apply this to $\mathfrak{g}_{\text{dR}}$ to obtain via PBW that

$$\mathcal{U}(\mathfrak{g}_{\text{dR}}) \cong \mathcal{U}(\mathfrak{g}) \otimes_{\mathbb{C}} \text{Sym}(\mathfrak{g}[1])$$

as $\mathcal{U}(\mathfrak{g})$ -modules. This proves it. ■

Since $\mathcal{U}(\mathfrak{g}_{\text{dR}}) \simeq \mathbb{C}$, we have a free resolution as a dg Lie algebra. Hence

$$\text{Ext}_{\mathcal{U}(\mathfrak{g})}^i(\mathbb{C}, M) \cong H^i(\text{Mod}_{\mathcal{U}(\mathfrak{g})}(\mathcal{U}(\mathfrak{g}_{\text{dR}}), M).$$

If A is a free $\mathbb{C}$ -algebra, then $\text{Mod}_A(A \otimes F, M) \cong \text{Mod}_{\mathbb{C}}(F, M)$. Applying this here yields

$$\text{Mod}_{\mathcal{U}(\mathfrak{g})}(\mathcal{U}(\mathfrak{g}_{\text{dR}}), M) \cong \text{Mod}_{\mathcal{U}(\mathfrak{g})}(\mathcal{U}(\mathfrak{g}) \otimes_{\mathbb{C}} \text{Sym}(\mathfrak{g}[1]), M) \cong \text{Vect}_{\mathbb{C}}(\text{Sym}(\mathfrak{g}[1]), M).$$

We often call this complex $C^\bullet(\mathfrak{g}; M)$. In degree 0, then this is $\text{Vect}_{\mathbb{C}}(\mathbb{C}, M) \cong M$. In degree 1, we take the degree -1 generators $\text{Vect}_{\mathbb{C}}(\mathfrak{g}, M)$. In degree 2, its $\text{Vect}_{\mathbb{C}}(\Lambda^2 \mathfrak{g}, M)$. This generalizes.

$$C^i(\mathfrak{g}; M) = \text{Vect}_{\mathbb{C}}(\Lambda^i \mathfrak{g}, M).$$

Cheveleakslky-Eilenberg's theorem si that if G is a connected compact Lie group, then the cohomology of the invariants of the de Rham complex $\Omega(G)^G$ are that of $C^\bullet(\mathfrak{g}; \mathbb{C})$. More powerfully, $\Omega(G)^G \cong C^\bullet(\mathfrak{g}; \mathbb{C})$. This follows because your tangent bundle for Lie groups is trivial so ur forms are constant coefficient. Hence we obtain a differential on $C^\bullet(\mathfrak{g}; \mathbb{C})$ via the exterior derivative.

We claim for any M , then $C^\bullet(\mathfrak{g}; M)$ is a $C^\bullet(\mathfrak{g}; \mathbb{C})$ module. Take $C^\bullet(\mathfrak{g}; M) \cong C^\bullet(\mathfrak{g}) \otimes M$. We may now define the differential. For trivial coefficients, d being a derivation implies we need only define it on generators:

$$C^\bullet(\mathfrak{g}) \cong \text{Sym}(\mathfrak{g}[1]^*) = \Lambda^\bullet \mathfrak{g}^*$$

has it as $0 : \mathbb{C} \rightarrow \mathfrak{g}$, and it suffices now only to define it on $\mathfrak{g}^* \rightarrow \Lambda^2 \mathfrak{g}^*$ since all others are generated on it. We define this as the adjunct of $[-, -] : \Lambda^2 \mathfrak{g} \rightarrow \mathfrak{g}$. Check wikipedia for a full explicit definition.

For $C^\bullet(\mathfrak{g}; M)$, we only need it on degree 0 since it is a $C^\bullet(\mathfrak{g})$ -module. The map $M \rightarrow \text{Hom}(\mathfrak{g}, M)$ is given by $m \mapsto (x \mapsto xm)$. Wikipedia also computes the first few cohomology groups. H2 corresponds to “central extensions” modulo some sort of isomorphism classes. These are extensions of $\mathfrak{g}$ as a Lie algebra by $\mathbb{C}$, meaning Lie algebras $\mathfrak{h}$ which extend $\mathfrak{g}$ in the sense taht

$$0 \longrightarrow \mathbb{C} \longrightarrow \mathfrak{h} \longrightarrow \mathfrak{g} \longrightarrow 0$$

is exact. Short exact sequences of vector spaces always split, so $\mathfrak{h} \cong \mathfrak{g} \oplus \mathbb{C}$. Thus its bracket has to be $[x, y] + \varphi(x, y)c$ for some basis $c \in \mathbb{C}$ and some $\varphi : \Lambda^2 \mathfrak{g} \rightarrow \mathbb{C}$. What in all of the fucks is happening???? review these notes later. see wikipedia page on Lie group cohomology, nLab page on dg Lie algebras, [CG16] A.3, and check the paper they cite at the end and apparently Lurie treats this for ∞ -categories somewhere so maybe I can find that.

Lecture 9: sdfgh

Fri 20 Feb 2026 14:01

Now we want to return to our goal of constructing $\mathcal{U}(\mathfrak{g})$ via prefactorization algebras. Might be useful for reviewing notes: $\mathcal{U}$ adjoint implies it commutes with $\oplus$, so $\mathcal{U}(\mathfrak{g})$ is graded if $\mathfrak{g}$ is graded.

Recall the *homology* is computed by $C^\bullet(\mathfrak{g}, M) \cong \text{Sym}(\mathfrak{g}[1]) \otimes M$ (dualize $\mathfrak{g}$ and $1 \rightarrow -1$ for cohomology), with differential $d_{\text{CE}} + d_{\mathfrak{g}} + d_M$, which acts by (up to the grading signs)

$$\begin{aligned} d(x_1 \cdot x_2 \cdots x_r \cdot m) &= \sum (-1)^? [x_i, x_j] x_1 \cdots \hat{x}_i \cdots \hat{x}_j \cdots x_r m \\ &\quad + (-1)^? \sum_{i=1}^n x_1 \cdots \hat{x}_i \cdots x_r (x_i m) \\ &\quad + (-1)^? \sum_{i=1}^r x_1 \cdots d_{\mathfrak{g}}(x_i) \cdots x_r m \\ &\quad + (-1)^? x_1 \cdots x_r d_M(m). \end{aligned}$$

Remark 3.1.14. Replaing normal commutativity with Kozkul's sign rulle yields

$$\text{Sym}(V) = \mathbb{T}(V) / \langle x \otimes y - (-1)^{|x||y|} y \otimes x \rangle.$$

Here $\mathbb{T}(V)$ is the tensor algebra.

For example, if V is concentrated in odd degree, then it becomes the exterior algebra. If concentrated in even degree, we get the usual symmetric algebra. The gading takes care of all relevant variations on the rules.

Definition 3.1.15. A differential graded algebra (DGA), (A, d_A) is a graded algebra such that

$$d_A(a \cdot b) = d_A(a)b + (-1)^{|a|} a \cdot d_A(b).$$

For example, de Rham's complex. In fact, de Rham's complex is *commutative* in the graded sense (sometimes called supercommutative???) meaning it is commutative up to Kozkul's sign rule.

Typically we will have $M = \mathbb{C}$. Then we write $C^\bullet(\mathfrak{g})$ and $C_\bullet(\mathfrak{g})$ for simplicity, and note that d_M has no contribution to the boundary morphism.

Warning 3.1.16. Although *cohomology* $C^\bullet(\mathfrak{g})$ is a DGA, *homology* $C_\bullet(\mathfrak{g})$ is NOT a DGA. It would be a "differential graded coalgebra".

Definition 3.1.17. Let $(\mathfrak{g}, d_{\mathfrak{g}})$ a dg Lie algebra. A Maurev-Cartan element in $\mathfrak{g}$ is some degree 1 $\alpha \in \mathfrak{g}_1$ such that

$$d_{\mathfrak{g}}(\alpha) + \frac{1}{2}[\alpha, \alpha] = 0.$$

Some kind of proof exists that like formal neighborhoods of a space in some moduli space are equivalent as ∞ -categories to some category of dg Lie algebras. In field theory, we work with the moduli space $EL(S)$ of solutions to the Euler Lagrange equations. Hence we can locally model these solutions, up to ∞ -categorical equivalence, by dg Lie algebras via taking the Maurev-Cartan locus or smt.

Proposition 3.1.18. *Let $(\mathfrak{g}, d_{\mathfrak{g}})$ a dgla. Then the (not Lie!) cohomology $H^{\bullet}(\mathfrak{g})$ is a graded Lie algebra.*

Proof. Let $\alpha \in H^i(\mathfrak{g})$ and $\beta \in H^j(\mathfrak{g})$. Then $[\alpha, \beta]$ by choosing a representative, computing the Lie bracket, and passing to cohomology. Note that the choice of representatives are closed (kernel mod image so everything in kernel). We need to check that the bracket of representatives is closed. This is immediate:

$$d[\tilde{\alpha}, \tilde{\beta}] = [d\tilde{\alpha}, \tilde{\beta}] + (-1)^{|\tilde{\alpha}|}[\tilde{\alpha}, d\tilde{\beta}] = [0, \tilde{\beta}] + (-1)^{|\alpha|}[\tilde{\alpha}, 0] = 0.$$

Now we shw it is well-defined. This is easy to check using that the elements r closed. ■

Let $U \subseteq X = \mathbb{R}$. Consider the precosheaf of dglas

$$U \mapsto (\mathfrak{g} \otimes \Omega_c^{\bullet}(U), d_{dR})$$

given by the compactly supported de Rham complex. Call this $\mathfrak{g}^{\mathbb{R}}(U)$. Ordinary de Rham cohomology is singular cohomology, and $\mathbb{R}^n$ is contractible so $H^i(\mathbb{R}) = \delta_{i0}\mathbb{R}$. Unfortunately, compactly supported cohomology is *not* homotopy invariant, so it turns out we will get

$$H_c^i(\mathbb{R}^n) = \delta_{in}\mathbb{R}.$$

Since $\mathfrak{g}^{\mathbb{R}}$ is just a cosheaf, we get

$$\mathfrak{g}^{\mathbb{R}}(U_1 \amalg U_2) \cong \mathfrak{g}^{\mathbb{R}}(U_1) \oplus \mathfrak{g}^{\mathbb{R}}(U_2).$$

If we take the symmetric algebra on that, it becomes a tensor product. So $U \mapsto C_{\bullet}(\mathfrak{g}^{\mathbb{R}}(U))$ is a prefactorization algebra in cochain complexes.

Proposition 3.1.19 (3.4.1). *Let $\mathcal{H}$ denote the cohomology pFA of $C_{\bullet}(\mathfrak{g}^{\mathbb{R}})$ by assigning*

$$\mathcal{H}(U) = H^{\bullet}(C_{\bullet}(\mathfrak{g}^{\mathbb{R}}(U))).$$

This is a pFA valued in graded vector spaces. Then $\mathcal{H}$ is locally constant (corresponding to a graded associative algebra). The corresponding graded algebra is isomorphic to $\mathcal{U}(\mathfrak{g})$.

Proof. Let $I \hookrightarrow J$ an inclusion of open intervals. Then there is a quasi-isomorphism $\mathfrak{g} \otimes \Omega_c^{\bullet}(I) \simeq \mathfrak{g} \otimes \Omega_c^{\bullet}(J)$. Note that quasi-isomorphic dglas have quasi-isomorphic chevelay-eilenberg complexes (simple lemma). In particular the functor $\mathfrak{g} \mapsto C_{\bullet}(\mathfrak{g})$ is a homotopic functor. Hence there is $C_{\bullet}(\mathfrak{g}^{\mathbb{R}}(I)) \simeq C_{\bullet}(\mathfrak{g}^{\mathbb{R}}(J))$, yielding an isomorphism in cohomology. ■

Lecture 10: More constructing $\mathcal{U}(g)$

Fri 27 Feb 2026 14:01

The complex $\mathfrak{g}^{\mathbb{R}}$ is quasi-isomorphic to the abelian dgla consisting of $\mathfrak{g}$ in degree 1, and 0 elsewhere. Since the differential is 0 and the bracket is 0 (thats what we mean by abelian), both $d_{\mathfrak{g}}$ and d_{CE} are 0. Hence

$$C_{\bullet}(\mathfrak{g}^{\mathbb{R}}(U)) \cong \text{Sym}(\mathfrak{g}[-1][1], d=0) = \text{Sym}(\mathfrak{g}).$$

Since $\mathfrak{g}$ lives in degree 0, this is a true symmetric algebra, giving us an associative algebra, which we call $A_{\mathfrak{g}}$. PBW says there is an increasing filtration on $\mathcal{U}(\mathfrak{g})$ and multiplication preserves this filtration. Hence $\mathcal{U}(\mathfrak{g}) \cong \text{Sym}(\mathfrak{g})$ as vector spaces. We now only check the multiplicative structure. We construct a linear isomorphism $\phi : \mathfrak{g} \rightarrow A_{\mathfrak{g}}$ and check that it is a map of Lie algebras, using the commutator for $A_{\mathfrak{g}}$. Transposing under $\mathcal{U} \dashv \text{Lie}$ yields a map of *algebras* $\mathcal{U}(\mathfrak{g}) \rightarrow A_{\mathfrak{g}}$.

Pick a bump function ε_i centered at $i \in \{\pm 1, 0\}$ with support $B_\delta(i)$, $\delta \ll \frac{1}{2}$, and integral 1. Let ε any compactly supported bump function with $\int_{\mathbb{R}} \varepsilon dx = 1$. Define $\phi : \mathfrak{g} \rightarrow A_{\mathfrak{g}} \cong H^\bullet(C_\bullet(\mathfrak{g}^{\mathbb{R}}(\mathbb{R})))$ by $X \mapsto [X \otimes \varepsilon dx]$. Since there are no brackets to take and since it is closed, this is a well-defined cohomology class under the differential $d_{\text{CE}} + d_{\text{dR}}$.

Recall multiplication in $A_{\mathfrak{g}}$ is done by choosing disjoint intervals I, J . Pick $\alpha, \beta \in A_{\mathfrak{g}}$, hence $\alpha \in \mathcal{H}(I), \beta \in \mathcal{H}(J)$. Then $\alpha\beta$ is the image of $\alpha \otimes \beta$ under the factorization structure map $\mathcal{H}(I) \otimes \mathcal{H}(J) \rightarrow \mathcal{H}(\mathbb{R}) = A_{\mathfrak{g}}$.

Now pick $X, Y \in \mathfrak{g}$. Then we pick compactly supported bump functions to write $\phi(X)\phi(Y) = [X \otimes \varepsilon_0 dx][Y \otimes \varepsilon_1 dx]$. Note that we choose the element multiplied on the left to be supported in an interval on the left. Thus $\phi(Y)\phi(X) = [Y \otimes \varepsilon_{-1} dx][X \otimes \varepsilon_0 dx]$. We also have

$$\phi[X, Y] = [[X, Y] \otimes \varepsilon_0 dx].$$

We show that ϕ preserves the bracket: in $\mathcal{H}(\mathbb{R})$,

$$\phi(X)\phi(Y) - \phi(Y)\phi(X) = \phi[X, Y].$$

Remark 3.1.20. Since the CE *chain* complex is not a dg algebra, meaning the differential is not compatible with the multiplicative structure, just because multiplication commutes in the symmetric algebra does not mean that multiplication is the same in cohomology. Hence multiplication of cohomology classes is not cohomology of multiplication.

Note that

$$\int_{\mathbb{R}} \varepsilon_{-1} dx - \varepsilon_1 dx = 0$$

since they are compactly supported. Then there is $h \in \Omega_c^0(\mathbb{R})$ with $dh = \varepsilon_{-1} dx - \varepsilon_1 dx$, since in $\mathbb{R}$ exact forms are precisely $\text{Ker } \int$ since $\mathbb{R} \simeq *$. Now consider

$$(d_{\text{CE}} + d_{\text{dR}})(X \otimes \varepsilon_0 dx)(Y \otimes h).$$

I think this is the product minus the bracket under ϕ . The CE differential gives $[X, Y] \otimes h \varepsilon_0 dx$, and sufficiently close to 0 we have $h \varepsilon_0 = \varepsilon_0$. Finally, de Rham kills $\varepsilon_0 dx$ and maps $h \mapsto dh$, so we get

$$= [X, Y] \otimes \varepsilon_0 dx + (X \otimes \varepsilon_0 dx)(Y \otimes dh) = [X, Y] \otimes \varepsilon_0 dx + (X \otimes \varepsilon_0 dx)(Y \otimes \varepsilon_1 dx - Y \otimes \varepsilon_{-1} dx).$$

Working out the algebra and using the fact that we may interchange things in Sym , this reduces to $\phi[X, Y] - \phi(X)\phi(Y) - \phi(Y)\phi(X)$. Since this is in the image of the differential, it vanishes in cohomology, so we are done.

Remark 3.1.21. Although the 1-forms are in degree 1 in de Rham, CE shifts degree by 1 so these objects do indeed commute, rather than anticommute.

3.2 (Pre)factorization envelopes

A sheaf $\mathcal{F}$ on a smooth manifold M is *fine* if it admits a partition of unity; i.e. for a locally finite cover $\{U_i\}_{i \in I}$ of M , there exist endomorphisms $\{\psi_i\}_{i \in I}$ of $\mathcal{F}$ such that $\sum_i \psi_i = 1$ and each ψ_i is supported in U_i . In particular, for any section $s \in \Gamma(U, \mathcal{F})$ there is a collection of sections $s_i \in U_i$ supported in U_i with $s = \sum_i s_i = \sum_i \psi_i(s)$ (?). For example, all smooth vector bundles over a smooth manifold.

Suppose we have a fine sheaf $\mathcal{L}$ on M valued in dglas , and $\mathcal{L}_c$ the cosheaf of its compactly supported sections. Fine-ness implies that $\mathcal{L}_c(U_1 \amalg U_2) = \mathcal{L}_c(U_1) \oplus \mathcal{L}_c(U_2)$. Define the pFA $\mathbb{U}\mathcal{L}$ as follows: for all open $V \subseteq M$, define

$$\mathbb{U}\mathcal{L} := C_\bullet(\mathcal{L}_c(V)).$$

Since Sym coherently takes $\oplus \rightarrow \otimes$, this is a preFA.

Examples: de Rham - $\mathcal{L}_{M,\mathfrak{g}}^{\text{dR}} = (\mathfrak{g} \otimes \Omega^\bullet(M), d_{\text{dR}})$. Dolbeaux (X complex manifold) - $\mathcal{L}_{X,\mathfrak{g}}^{\text{Dol}} = (\mathfrak{g} \otimes \Omega_X^{0,\bullet}, \bar{\partial}_X)$ (holomorphic envelope but not holomorphic sections - those aren't fine).

Recall in physics that we care about states up to global phase, and in particular up to scaling by normalization. Hence physics is really a projective space. Let $\mathfrak{g}$ a Lie algebra and $\mathfrak{h} \subseteq \mathfrak{g}$ a (Lie) ideal. For example, the center. The center of $\mathfrak{gl}(n, \mathbb{C})$, for example, is $\mathbb{C}1$. We can quotient by ideals in LieAlg .

A *central extension* of a Lie algebra $\mathfrak{g}$ is a Lie algebra $\hat{\mathfrak{g}}$ such that $\mathfrak{g} \cong \hat{\mathfrak{g}}/\mathfrak{a}$, for $\mathfrak{a}$ an abelian ideal contained in the center $\mathfrak{a} \subseteq Z(\hat{\mathfrak{g}})$. Equivalently, it is a short exact sequence

$$0 \longrightarrow \mathfrak{g} \longrightarrow \hat{\mathfrak{g}} \longrightarrow \mathfrak{a} \longrightarrow 0$$

with $\mathfrak{a}$ an abelian ideal.

Claim 3.2.1. *Central extensions of $\mathfrak{g}$ by $\mathfrak{a}$ are classified by a certain Lie algebra cohomology group $H^2(\mathfrak{g}, \mathfrak{a})$, where $\mathfrak{a}$ is the trivial module. I.e., every central extension of $\mathfrak{g}$ is described by a 2-cocycle $\Lambda^2 \mathfrak{g} \rightarrow \mathfrak{a}$.*

Proof. Consider a central extension

$$0 \longrightarrow \mathfrak{g} \longrightarrow \hat{\mathfrak{g}} \longrightarrow \mathfrak{a} \longrightarrow 0.$$

Choose a split as vector spaces (i.e. a section j of the projection). Let $\omega \in \Lambda^2 \mathfrak{g}^* \otimes \mathfrak{a} = \text{Hom}(\Lambda^2 \mathfrak{g}, \mathfrak{a})$ as

$$\omega(X, Y) = [j(X), j(Y)] - j[X, Y].$$

If j were a Lie algebra homomorphism, $\omega = 0$. But it need not be. We can show this cocycle vanishes in cohomology iff j is a split as a sequence of Lie algebras.

For the converse, let $\omega : \Lambda^2 \mathfrak{g} \rightarrow \mathfrak{a}$ which dies in d_{CE} . Define $\hat{\mathfrak{g}}$ as $\mathfrak{g} \oplus \mathfrak{a}$ as vector spaces, with bracket

$$[(x, a), (y, b)] = ([x, y], \omega(x, y)).$$

■

Lecture 11: Weibel

Sat 28 Feb 2026 21:27

Lie algebras

To improve my understanding of dg Lie algebra cohomology, I read Weibel chapter 7, which focuses on Lie algebra cohomology. We begin with some basic notions.

Let R be a commutative ring. A *Lie algebra* $\mathfrak{g}$ is a R -module together with a *Lie bracket* $[-, -] : \mathfrak{g} \otimes_R \mathfrak{g} \rightarrow \mathfrak{g}$ which is not necessarily associative, but is antisymmetric $[x, x] = 0$ and satisfies the Jacobi identity

$$[x, [y, z]] + [y, [z, x]] + [z, [x, y]] = 0.$$

An *ideal* is a R -submodule $\mathfrak{h} \subseteq \mathfrak{g}$ such that $[\mathfrak{g}, \mathfrak{h}] \subseteq \mathfrak{h}$. An ideal is immediately a Lie subalgebra, and one can quotient by an ideal to obtain a Lie algebra. A morphism of Lie algebras is a map $\varphi : \mathfrak{g} \rightarrow \mathfrak{g}'$ which commutes with the bracket in the sense that $\varphi[-, -] = [\varphi(-), \varphi(-)]$. Every ideal can thus be written as a short exact sequence of Lie algebras

$$0 \longrightarrow \mathfrak{h} \longrightarrow \mathfrak{g} \longrightarrow \mathfrak{g}/\mathfrak{h} \longrightarrow 0.$$

An *abelian* Lie algebra is a Lie algebra where $[-, -] = 0$.

Intuitively, the bracket is a commutator, and indeed every associative R -algebra A can be viewed as a Lie algebra by defining $[x, y] = xy - yx$. We write $\text{Lie}(A)$ for this Lie algebra, and thus we obtain a functor $\text{Lie} : \text{Alg}_R \rightarrow \text{Lie}$. This functor has a left adjoint given by the universal enveloping algebra $U : \text{Lie} \rightarrow \text{Alg}_R$, which we will explore indepth later.

Given an associative R -algebra A , we obtain the Lie algebra $\mathfrak{gl}_n(A)$ of matrices with values in A , together with important Lie subalgebras

- $\mathfrak{sl}_n(A)$: the set of traceless matrices. This becomes difficult to define precisely when A is noncommutative.
- $\mathfrak{o}_n(A)$: the set of skew-symmetric matrices $A_{ij} = -A_{ji}$.
- $\mathfrak{t}_n(A)$: the set of upper triangular matrices.
- $\mathfrak{n}_n(A)$: the set of strictly upper triangular matrices.

Let A a not necessarily associative R -algebra. A derivation of A into itself is a R -linear map $D : A \rightarrow A$ satisfying a Leibnitz rule

$$D(ab) = (D(a))b + a(D(b)).$$

The set $\text{Der}(A)$ of derivations on A has a commutator $[D_1, D_2] = D_1 \circ D_2 - D_2 \circ D_1$, which makes $\text{Der}(A)$ into a Lie algebra.

Given an R -module M , the *free Lie algebra* $\mathfrak{f}(M)$ is a Lie algebra containing M as a submodule which is universal with respect to turning R -linear maps into Lie algebra maps. The free Lie algebra assembles into a functor $\mathfrak{f} : \text{Lie} \rightarrow \text{Mod}_R$ with $\mathfrak{f} \dashv U$ for U the forgetful functor. The existence of such a Lie algebra follows by the adjoint functor theorem.

Given Lie algebras $\mathfrak{g}, \mathfrak{h}$, the product Lie algebra $\mathfrak{g} \times \mathfrak{h}$ is defined by

$$[(g, h), (g', h')] = ([g, g'], [h, h']).$$

The *lower central sequence* of a Lie algebra is the descending sequence of ideals

$$\mathfrak{g} \supseteq \mathfrak{g}^2 = [\mathfrak{g}, \mathfrak{g}] \supseteq \mathfrak{g}^3 = [\mathfrak{g}^2, \mathfrak{g}] \supseteq \cdots \supseteq \mathfrak{g}^n = [\mathfrak{g}^{n-1}, \mathfrak{g}] \supseteq \cdots.$$

A Lie algebra is *nilpotent* if $\mathfrak{g}^n = 0$ for some n . For example, $\mathfrak{n}_n(A)$ is nilpotent, and so are abelian Lie algebras.

The *derived series* of a Lie algebra $\mathfrak{g}$ is the following descending sequence of ideals

$$\mathfrak{g} \supseteq \mathfrak{g}^{(1)} = [\mathfrak{g}, \mathfrak{g}] \supseteq \mathfrak{g}^{(2)} = [\mathfrak{g}^{(1)}, \mathfrak{g}^{(1)}] \supseteq \cdots \supseteq \mathfrak{g}^{(n)} = [\mathfrak{g}^{(n-1)}, \mathfrak{g}^{(n-1)}] \supseteq \cdots.$$

We say that $\mathfrak{g}$ is *solvable* if $\mathfrak{g}^{(n)} = 0$ for some n . For example, every nilpotent Lie algebra is solvable: simply show $[\mathfrak{g}^i, \mathfrak{g}^j] \subseteq \mathfrak{g}^{i+j}$ and the statement follows inductively. The converse is not true – for example, $\mathfrak{t}_n(A)$ is solvable but not nilpotent.

$\mathfrak{g}$ -Modules

Definition 3.2.2. A (left) $\mathfrak{g}$ -module M is an R -module together with a R -bilinear multiplication map $\mathfrak{g} \otimes_R M \rightarrow M$ (written $x \otimes m \mapsto xm$) such that

$$[x, y]m = x(ym) - y(xm).$$

A $\mathfrak{g}$ -module homomorphism is an R -linear map $\varphi : M \rightarrow N$ such that $\varphi(xm) = x\varphi(m)$. We write $\text{Hom}_{\mathfrak{g}}(M, N)$ or $\text{Mod}_{\mathfrak{g}}(M, N)$ for the set of $\mathfrak{g}$ -module homomorphisms, yielding an abelian category $\text{Mod}_{\mathfrak{g}}$.

Note that $\text{Hom}_{\mathfrak{g}}(M, N) \subseteq \text{Hom}_R(M, N)$ is a R -submodule.

There is a trivial $\mathfrak{g}$ -module functor from $\text{Mod}_R \rightarrow \text{Mod}_{\mathfrak{g}}$ given by taking an R -module M , and defining the action by $\mathfrak{g}$ to be the trivial action $0 : \mathfrak{g} \otimes_R M \rightarrow M$. It is an exact functor, and elements of its image are called *trivial $\mathfrak{g}$ -modules*. Write this functor as $\mathbf{0}$. It has left- and right-adjoints.

The *invariant submodule* $M^{\mathfrak{g}}$ of a $\mathfrak{g}$ -module M is the submodule

$$M^{\mathfrak{g}} = \{m \in M \mid xm = 0 \forall x \in \mathfrak{g}\}.$$

We note that $M^{\mathfrak{g}} \cong \text{Mod}_{\mathfrak{g}}(R, M)$, when viewing R as a trivial R -module. This follows since R -linear maps $\varphi : R \rightarrow M$ are determined by $\varphi(1)$ since $\varphi(r) = r\varphi(1)$. To make it $\mathfrak{g}$ -linear, we need $x\varphi(1) = \varphi(x1) = \varphi(0) = 0$ since R is a trivial module, hence we may only map 1 to an element of $M^{\mathfrak{g}}$. Moreover such a choice uniquely determines φ . Since it is the maximal trivial submodule of $M^{\mathfrak{g}}$, we obtain $\mathbf{0} \dashv (-)^{\mathfrak{g}}$. This is immediate to see: a $\mathfrak{g}$ -module homomorphism $\mathbf{0}(N) \rightarrow M$ must identify $x\varphi(n) = 0$, so its image must fall into a trivial submodule, hence it factors through $M^{\mathfrak{g}}$. Moreover, any R -linear map $N \rightarrow M^{\mathfrak{g}}$ induces a map $\mathbf{0}(N) \rightarrow M$ in this way. Hence we have $\mathbf{0} \dashv -^{\mathfrak{g}}$. Additionally, since $M^{\mathfrak{g}} \cong \text{Mod}_{\mathfrak{g}}(\mathbf{0}(R), M)$, we have a representation $-^{\mathfrak{g}} \cong \text{Mod}_{\mathfrak{g}}(\mathbf{0}(R), -)$.

The *coinvariants* of M are the quotient $M/\mathfrak{g}M$. Note that this is also a trivial module since for any $x \in \mathfrak{g}$ and $m \in M$, we have $xm \in \mathfrak{g}M$ which is identified to 0 in the quotient. Since this is the largest quotient which is a trivial module, we can show that $(-)^{\mathfrak{g}} \dashv \mathbf{0}$. We will see later that $\text{Mod}_{\mathfrak{g}}$ has enough projectives and injectives, hence we can compute the derived functors of (co)invariants.

Definition 3.2.3. Let M be a $\mathfrak{g}$ -module. We write $H_{\bullet}(\mathfrak{g}, M)$ or $H_{\bullet}^{\text{Lie}}(\mathfrak{g}, M)$ for the derived functors $L_{\bullet}(-)^{\mathfrak{g}}(M)$ of $-^{\mathfrak{g}}$, and call them the *homology groups of $\mathfrak{g}$ with coefficients in M* . By definition, $H_0(\mathfrak{g}, M) = M_{\mathfrak{g}}$.

Similarly, write $H^{\bullet}(\mathfrak{g}, M)$ or $H_{\text{Lie}}^{\bullet}(\mathfrak{g}, M)$ for the right derived functors $R^{\bullet}(-)^{\mathfrak{g}}(M)$ of $-^{\mathfrak{g}}$, and call them the *cohomology groups of $\mathfrak{g}$ with coefficients in M* . By definition, $H^0(\mathfrak{g}, M) = M^{\mathfrak{g}}$.

Universal Enveloping Algebras

Write $T(\mathfrak{g})$ for the tensor algebra on $\mathfrak{g}$. If $\mathfrak{g}$ is a Lie algebra, the universal enveloping algebra can be given as $T(\mathfrak{g})$ modulo the relation that the inclusion $\mathfrak{g} \hookrightarrow T(\mathfrak{g})$ be a Lie algebra homomorphism and $T(\mathfrak{g})$ have bracket given by the commutator. It is universal in the sense that precomposition by the inclusion yields

$$\text{Lie}(-, \text{Lie}) \cong \text{Alg}_R(U, -).$$

Proposition 3.2.4. *If $\mathfrak{g}$ is a Lie algebra, then every $\mathfrak{g}$ -module is naturally a $U\mathfrak{g}$ -module, and vice versa. This yields natural isomorphisms $\text{Mod}_{\mathfrak{g}} \cong \text{Mod}_{U\mathfrak{g}}$.*

Proof. A $\mathfrak{g}$ -module is a Lie algebra map $\mathfrak{g} \rightarrow \text{Lie}(\text{End}_R(M))$ for some R -module M . A $U\mathfrak{g}$ -module is an algebra map $U\mathfrak{g} \rightarrow \text{End}_R(M)$. The statement now follows by adjointness

$$\text{Lie}(\mathfrak{g}, \text{Lie}(\text{End}_R(M))) \cong \text{Alg}_R(U\mathfrak{g}, \text{End}_R(M)).$$

■

Since the categories of associative algebras have enough projectives and injectives, thus so do the categories of $\mathfrak{g}$ -modules. The action is simply given by, for $x_1, \dots, x_n \in \mathfrak{g}$,

$$(x_1 \cdots x_n)m = x_1(\cdots(x_nm)\cdots).$$

We then impose bilinearity to cover all of $U\mathfrak{g}$.

Corollary 3.2.5. *Let M be a $\mathfrak{g}$ -module. Then*

$$H_\bullet(\mathfrak{g}, M) \cong \text{Tor}_\bullet^{U\mathfrak{g}}(R, M),$$

$$H^\bullet(\mathfrak{g}, M) \cong \text{Ext}_{U\mathfrak{g}}^\bullet(R, M).$$

Proof. Since we have already observed that $M^\mathfrak{g} \cong \text{Mod}_\mathfrak{g}(R, -) \cong \text{Mod}_{U\mathfrak{g}}(R, -)$, the second statement is immediate by functoriality of derived functors (in the homotopy sense) and homology. For the first, note that the transpose of $0 : \mathfrak{g} \rightarrow R$ under $U \dashv \text{Lie}$ is the unique map $U\mathfrak{g} \rightarrow R$ mapping $\mathfrak{g} \mapsto 0$. Define the *augmentation ideal* $\mathfrak{J}$ to be the kernel of this map; evidently $\mathfrak{J}$ is the two-sided ideal generated by the image of $\mathfrak{g}$ in $U\mathfrak{g}$. This map is epi, hence the first isomorphism theorem yields $R \cong U\mathfrak{g}/\mathfrak{J} \cong (U\mathfrak{g})_\mathfrak{g}$. Then

$$R \otimes_{U\mathfrak{g}} M \cong (U\mathfrak{g}/\mathfrak{J}) \otimes_{U\mathfrak{g}} M \cong M/\mathfrak{J}M \cong M/\mathfrak{g}M \cong M_\mathfrak{g}.$$

■

Theorem 3.2.6 (Poincaré-Birkhoff-Witt). *Let $\mathfrak{g}$ be a free R -module. Then $U\mathfrak{g}$ is also a free R -module. If $\{e_\alpha\}_{\alpha \in A}$ is an ordered basis of $\mathfrak{g}$, then the collection of all elements of the form e_I , where I is an increasing sequence, form a basis for $U\mathfrak{g}$.*

We decline to prove the PBW.

Corollary 3.2.7. *The inclusion $\mathfrak{g} \hookrightarrow U\mathfrak{g}$ is injective, so we may identify $\mathfrak{g}$ with its image.*

Corollary 3.2.8. *If $\mathfrak{h} \subseteq \mathfrak{g}$ is a Lie subalgebra, and $R = k$ is a field, then $U\mathfrak{g}$ is a free $U\mathfrak{h}$ -module.*

Exercise 3.2.9. Let M, N be left $\mathfrak{g}$ -modules. Show that $\text{Mod}_R(M, N)$ is a $\mathfrak{g}$ -module by $(xf)(m) = xf(m) - f(xm)$. Then show that $\text{Mod}_\mathfrak{g}(M, N) \cong \text{Mod}_R(M, N)^\mathfrak{g}$. Show that this induces isomorphisms $\text{Ext}_{U\mathfrak{g}}^\bullet(M, N) \cong H_{\text{Lie}}^\bullet(\mathfrak{g}, \text{Mod}_R(M, N))$. In particular, the global dimension of $U\mathfrak{g}$ equals the Lie algebra cohomological dimension of $\mathfrak{g}$.

Proof. For the first, we need only show $[x, y]f = x(yf) - y(xf)$. Let $x, y \in \mathfrak{g}$ and $m \in M$. Then since M, N are $\mathfrak{g}$ -modules,

$$\begin{aligned} ([x, y]f)(m) &= [x, y]f(m) - f([x, y]m) \\ &= xyf(m) - yxf(m) - f(xym - yxm) \\ &= xyf(m) - yxf(m) - f(xym) + f(yxm). \end{aligned}$$

Note that $xf = x \circ f - f \circ x$. Thus we also have

$$x(yf) = x(y \circ f - f \circ y) = x \circ y \circ f - y \circ f \circ x - x \circ f \circ y + f \circ y \circ x.$$

Hence evaluating at m yields

$$(x(yf))(m) = xyf(m) - yf(xm) - xf(ym) + f(yxm).$$

Similarly,

$$(y(xf))(m) = yxf(m) - xf(ym) - yf(xm) + f(xym).$$

Thus

$$\begin{aligned} & (x(yf))(m) - (y(xf))(m) \\ &= xyf(m) - yf(xm) - xf(ym) + f(yxm) - yxf(m) + xf(ym) + yf(xm) - f(xym) \\ &= xyf(m) + f(yxm) - yxf(m) - f(xym). \end{aligned}$$

They agree, hence $\text{Mod}_R(M, N)$ is a $\mathfrak{g}$ -module in the way claimed.

Now let f be a $\mathfrak{g}$ -invariant of $\text{Mod}_R(M, N)$. Then $0 = xf = x \circ f - f \circ x$ for all $x \in \mathfrak{g}$, and thus $x \circ f = f \circ x$. Thus f is $\mathfrak{g}$ -linear. Conversely if f is $\mathfrak{g}$ -linear, then $f \circ x = x \circ f$, and thus $xf = x \circ f - f \circ x = 0$, so f is a $\mathfrak{g}$ -invariant of $\text{Mod}_R(M, N)$. ■

H^1 and H_1

Since $\mathfrak{J}$ is the kernel of $U\mathfrak{g} \rightarrow R$, there is a short exact sequence of $\mathfrak{g}$ -modules

$$0 \longrightarrow \mathfrak{J} \longrightarrow U\mathfrak{g} \longrightarrow R \longrightarrow 0.$$

Since $H_\bullet(\mathfrak{g}, -) \cong \text{Tor}_\bullet^{U\mathfrak{g}}(R, -)$, we apply the *dimension shifting technique* to take $H_n(\mathfrak{g}, -) = H_{n-1}(\mathfrak{g}, -)$: consider the long exact sequence

$$\cdots \longrightarrow \text{Tor}_n^{U\mathfrak{g}}(U\mathfrak{g}, M) \longrightarrow \text{Tor}_n^{U\mathfrak{g}}(R, M) \longrightarrow \text{Tor}_{n-1}^{U\mathfrak{g}}(\mathfrak{J}, M) \longrightarrow \text{Tor}_{n-1}^{U\mathfrak{g}}(U\mathfrak{g}, M) \longrightarrow \cdots.$$

Since $U\mathfrak{g}$ is free as a $U\mathfrak{g}$ -module, it is flat, and hence its Tor modules vanish. Thus the short exact sequence simply reads

$$\cdots \longrightarrow 0 \longrightarrow \text{Tor}_n^{U\mathfrak{g}}(R, M) \longrightarrow \text{Tor}_{n-1}^{U\mathfrak{g}}(\mathfrak{J}, M) \longrightarrow 0 \longrightarrow \cdots.$$

The only way for this to be exact is if the map $\text{Tor}_n^{U\mathfrak{g}}(R, M) \rightarrow \text{Tor}_{n-1}^{U\mathfrak{g}}(\mathfrak{J}, M)$ is an isomorphism for $n \geq 2$. Additionally, reading this in degree $n = 1, 0$, we obtain the exact sequence

$$0 \longrightarrow H_1(\mathfrak{g}, M) \longrightarrow \mathfrak{J} \otimes_{U\mathfrak{g}} M \longrightarrow M \longrightarrow M_{\mathfrak{g}} \longrightarrow 0.$$

Here we have noted that $R \otimes_{U\mathfrak{g}} M \cong M_{\mathfrak{g}}$.

Exercise 3.2.10. Show that the inclusion $i : \mathfrak{g} \rightarrow U\mathfrak{g}$ maps $[\mathfrak{g}, \mathfrak{g}]$ to $\mathfrak{J}^2$. Conclude that it induces a map $i : \mathfrak{g}^{ab} \rightarrow \mathfrak{J}/\mathfrak{J}^2$, where $\mathfrak{g}^{ab} = \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$.

Proof. We consider the map $0 : \mathfrak{g} \rightarrow R$. ■

Lecture 12: Prefactorization envelopes (pFE's)

Tue 03 Mar 2026 14:01

Let $\mathcal{L}$ be a fine sheaf on a manifold M of dgla's, and $\mathcal{L}_c$ the corresponding cosheaf of compactly supported sections. The factorization envelope $\mathbb{U}\mathcal{L}$ is defined as the prefactorization algebra mapping $U \subseteq M$ to Chevalley chains $C_\bullet(\mathcal{L}_c(U))$. This is a preFA in cochain complexes, and the cohomology of this (which is the dgla homology) yields a preFA $H^\bullet(C_\bullet(\mathcal{L}_c(U)))$ in vector spaces.

It's often useful to consider *twisted* variants of this construction corresponding to central extensions of $\mathcal{L}$. Recall given a Lie algebra $\mathfrak{g}$, a central extension of $\mathfrak{g}$ is a Lie algebra $\hat{\mathfrak{g}}$ with a short exact sequence

$$0 \longrightarrow \mathfrak{a} \longrightarrow \hat{\mathfrak{g}} \longrightarrow \mathfrak{g} \longrightarrow 0,$$

where $\mathfrak{a} \subseteq Z(\hat{\mathfrak{g}})$ is an ideal contained inside the center. There is a correspondence of central extensions with $H^2(\mathfrak{g}, \mathfrak{a})$. Given a splitting of this sequence $j : \mathfrak{g} \rightarrow \hat{\mathfrak{g}}$, we define $\omega(x, y) = [j(x), j(y)] - j([x, y])$. We can show $d_{\text{CE}}\omega = 0$. We can also show that if we choose a different splitting, then ω changes only by a coboundary. Write another split as $j' = j + \alpha$, then

$$\omega'(x, y) = [j(x) + \alpha(x), j(y) + \alpha(y)] - j[x, y] - \alpha[x, y].$$

We must have $\alpha : \mathfrak{g} \rightarrow \mathfrak{a}$, so $\alpha(x), \alpha(y)$ fall in the center and don't affect the bracket. So it is just

$$\omega' = [j, j] - j[\cdot, \cdot] - \alpha[\cdot, \cdot].$$

We see then that α is a coboundary since $x \wedge y \rightarrow \alpha[x, y]$ is $d_{\text{CE}}\alpha$ in $C^\bullet(\mathfrak{g}, \mathfrak{a})$ apparently. So the cohomology class of ω is well-defined.

Example 3.2.11. Let $\mathfrak{g} = \mathbb{C}^2$ an abelian Lie algebra, and consider the 1-dimensional central extensions

$$0 \longrightarrow \mathbb{C} \longrightarrow \hat{\mathfrak{g}} \longrightarrow \mathbb{C}^2 \longrightarrow 0.$$

Then $H^2(\mathfrak{g}, \mathbb{C})$ looks like

$$0 \longrightarrow \mathfrak{g}^* \longrightarrow \Lambda^2 \mathfrak{g}^* \longrightarrow \Lambda^3 \mathfrak{g}^* \longrightarrow \dots$$

Given $\phi \in \Lambda^n \mathfrak{g}^*$ and $\{x_i\}_{1 \leq i \leq n+1} \subseteq \mathfrak{g}$, the differential is

$$(d_{\text{CE}}\phi)(x_1 \wedge \dots \wedge x_{n+1}) = \sum_{i,j} \pm \phi([x_i, x_j], x_1, \dots, \hat{x}_i, \dots, \hat{x}_j, \dots, x_n).$$

The $\pm$ is determined by the parity of i, j . Since the brackets are all 0, $d_{\text{CE}}\phi = 0$, so the homology is simply $H^2(\mathfrak{g}, \mathbb{C}) \cong \Lambda^2 \mathfrak{g}^*$. Since this is 1-dimensional over $\mathbb{C}$ since $\mathfrak{g}^*$ is 2-dimensional, there is a unique central extension up to scaling. Thus $\hat{\mathfrak{g}} \cong \mathfrak{g} \oplus \mathbb{C}k$ for some $k \in \mathbb{C}$, and $\omega(x, y) = k$ (here x, y the coordinates of $\mathbb{C}^2$). There is then one bracket $[x, y] = k$. This is the *Heisenberg Lie algebra* in 3 dimensions. These can be thought of as upper-triangular 3 dimensional matrices of the form

$$\begin{pmatrix} 0 & x & k \\ 0 & 0 & y \\ 0 & 0 & 0 \end{pmatrix}.$$

Example 3.2.12 (Affine Ka(c?z?idk)-Moody algebras). Let $\mathfrak{g}$ be a simple Lie $\mathbb{C}$ -algebra. If A is a commutative $\mathbb{C}$ -algebra, then $\mathfrak{g} \otimes_{\mathbb{C}} A$ is a Lie algebra $[x \otimes a, y \otimes b] = [x, y] \otimes ab$. This may fail

in noncommutative rings due to failiure of Jacobi's identity. Take A either Laurant polynomials $\mathbb{C}[t^{-1}, t]$ or Laurent series $\mathbb{C}((t))$. In the first case, the *Loop algebra* is defined as

$$L\mathfrak{g} = \mathfrak{g} \otimes \mathbb{C}[t^{-1}, t] \cong \mathfrak{g}[t^{-1}, t].$$

If we identify t with $e^{i\theta}$ ($\mathbb{C}$ -polynomials in z are polynomials in $e^{i\theta}$), then we may think of this as $C^\infty(S^1, \mathfrak{g})$, justifying the name.

Consider the cocycle $\langle -, - \rangle_{\mathfrak{g}}$, a symmetric $\mathfrak{g}$ -invariant form (so all multiples of the killing form). We will define

$$\omega(x \otimes f(t), y \otimes g(t)) = \langle x, y \rangle_{\mathfrak{g}} \operatorname{Res}_{t=0} f dg.$$

Skew-symmetry follows since the residue of $f dg$ is negative of that of $g df$; easy to check. We claim $\omega \in H^2(\mathfrak{g}, \mathbb{C})$, and

$$\hat{\mathfrak{g}} = L\mathfrak{g} \otimes \mathbb{C}k,$$

$$[x \otimes f(t), y \otimes g(t)] = [x, y] \otimes f(t)g(t) + \omega(x \otimes f, y \otimes g)k.$$

Example 3.2.13 (Virasoro algebra). Consider the Witt algebra of algebraic vector fields on $\mathbb{C}^*$; equivalently as vector fields on S^1 . This is $\mathbb{C}[t, t^{-1}]\partial_t$. Then

$$[f\partial_t, g\partial_t] = (fg' - gf')\partial_t.$$

We present the Witt algebra as span $\{L_n = -t^{n+1}\partial_t\}$ with

$$[L_m, L_n] = (m - n)L_{m+n}.$$

Then $H^2(\text{Witt}, \mathbb{C}) \cong \mathbb{C}$ has an interesting cocycle

$$\omega(L_m, L_n) = \frac{1}{12}(m^2 - m)\delta_{m(-n)}.$$

In string theory, we consider things called world sheets, and we are interested in gluing their boundaries (eg. spheres). Note that $\text{Lie}(\text{Diff}(S^1)) \cong \mathfrak{X}(S^1)$ – this should hold for all manifolds. The Witt algebra should act on this since it looks similar. QFT takes place in a complex Hilbert space $\mathcal{H}$, but the physical quantities only depend on the projective space $\mathbb{P}(\mathcal{H})$ since they don't depend on global phase. So the symmetries of a quantum system are actually $\text{Aut}(\mathbb{P}(\mathcal{H})) =: \text{PGL}(\mathcal{H})$, and we can write this as a central extension of groups

$$0 \longrightarrow \mathbb{C}^* \longrightarrow \text{GL}(\mathcal{H}) \longrightarrow \text{PGL}(\mathcal{H}) \longrightarrow 0.$$

Differentiating this yields a central extension of Lie algebras. A symmetry group G on a QFT is then a morphism $G \rightarrow \text{PGL}(\mathcal{H})$. Since projective geometry is harder than working in $\mathcal{H}$, we would like to lift $G \rightarrow \text{PGL}(\mathcal{H})$ to $\text{GL}(\mathcal{H})$ as indicated by the dotted arrow:

$$\begin{array}{ccccccc} 0 & \longrightarrow & \mathbb{C}^* & \longrightarrow & \text{GL}(\mathcal{H}) & \longrightarrow & \text{PGL}(\mathcal{H}) \longrightarrow 0 \\ & & \parallel & & \uparrow & \nearrow & \uparrow \\ 0 & \dashrightarrow & \mathbb{C}^* & \dashrightarrow & \hat{G} & \dashrightarrow & G \dashrightarrow 0 \end{array}$$

This dotted arrow need not always exist, but we can always find the dashed arrows to obtain a central extension, allowing us to consider $\hat{G}$ instead. Differentiating this does the same thing for Lie algebras ($d = T_1$). Interesting paper by Dan Freed about stuff like this for mroe physics and it uses categories.

Ok back to pFE's. Let $\mathcal{L} : \mathcal{O}pen(M)^{op} \rightarrow \mathcal{DGLie}$ a fine sheaf. An r -shifted central extension of $\mathcal{L}_c$ is a cosheaf of dgla's $\widehat{\mathcal{L}}_c$ lifting into an exact sequence of dgla's

$$0 \longrightarrow \mathbb{C}[r] \longrightarrow \widehat{\mathcal{L}}_c \longrightarrow \mathcal{L}_c \longrightarrow 0.$$

In this case we can form the *twisted envelope* $\mathbb{U}\widetilde{\mathcal{L}} : U \rightarrow C_\bullet(\widehat{\mathcal{L}}_c(U))$.

For example, consider

$$\mathcal{L} = \mathfrak{g} \otimes \Omega_{\mathbb{R}}^\bullet = \mathfrak{g}^{\mathbb{R}}.$$

Pick a cocycle $H^2(\mathfrak{g}, \mathbb{C}) \ni \omega : \mathfrak{g} \otimes \mathfrak{g} \rightarrow \mathbb{C}$ and write

$$\widehat{\mathcal{L}} = \mathfrak{g} \otimes \Omega_{\mathbb{R}}^\bullet \oplus \mathbb{C}[?]k.$$

To figure out what the shift is by, consider the thing we want the bracket to be

$$[x \otimes \alpha, y \otimes \beta] = [x, y] \otimes (\alpha \wedge \beta) + \left(\int_{\mathbb{R}} \omega(x, y) \alpha \wedge \beta \right) k.$$

One simply figures out what order everything has to be and determines that k must be degree 1, so we shift -1 .

Bibliography

- [CG16] Kevin Costello and Owen Gwilliam. *Factorization Algebras in Quantum Field Theory Volume 1*. Cambridge University Press, New Mathematical Monographs, 2016.
- [CG20] Kevin Costello and Owen Gwilliam. *Factorization Algebras in Quantum Field Theory Volume 2*. Cambridge University Press, New Mathematical Monographs, 2020.
- [Lee00] John Lee. *Smooth Manifolds*. Springer, 2000.